

Detection and Resolution of Clickbait Headlines in Russian Digital Media: Corpus Construction and Comparative Evaluation of Language Models

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Abstract: We present a Russian-language corpus for clickbait characterisation and resolution, together with an empirical evaluation of current-generation Large Language Models on the resulting resource. Building on the methodology of Noticia (García-Ferrero and Altuna, 2024), we construct a corpus of 1,296 Russian news articles collected from four digital outlets (eg.ru, aif.ru, lenta.ru, mail.ru) through comprehensive, undifferentiated web crawling rather than pre-curated clickbait sourcing, of which 600 are annotated as clickbait using an original, empirically-calibrated ten-feature taxonomy and severity-scoring formula, and all 1,296 carry a human-written one-sentence answer summary. We evaluate four general-purpose LLMs under a unified few-shot protocol and report an exploratory cross-lingual fine-tuning experiment adapting ClickbaitFighter-7B to Russian.

Keywords: Clickbait, Summarization, Dataset, Russian, Large Language Models.

1. Introducción

1.1 Background and Relevance of the Clickbait Phenomenon

The digitalisation of news consumption has fundamentally altered the economic incentives that govern journalistic production. In the attention economy that characterises contemporary digital media, advertising revenue is overwhelmingly tied to metrics of engagement — page views, click-through rates, and time spent on site — rather than to the intrinsic informational value of a given piece of content. This economic architecture has given rise to *clickbait*: headlines deliberately engineered to maximise the likelihood of a click, frequently at the expense of accuracy, completeness, or journalistic integrity. As García-Ferrero and Altuna (2024) note in their foundational work on the Noticia dataset, clickbait headlines exaggerate and distort the actual content of an article, exploiting an epistemic gap between what the reader is promised and what the reader actually needs to know, in order to compel a click that may leave the underlying informational need unmet (García-Ferrero & Altuna, 2024).

Clickbait headlines exploit a well-documented psychological mechanism known in the curiosity literature as the *information-gap theory* (Loewenstein, 1994): when an individual perceives a gap between what they know and what they want to know, this gap is experienced as an aversive state of deprivation that motivates information-seeking behaviour. Clickbait authors deliberately manufacture such gaps — through vague pronouns, incomplete comparisons, withheld identities, or unresolved rhetorical questions — knowing that the

resulting curiosity will translate into measurable engagement. This mechanism explains why clickbait remains a persistent and, from a purely commercial standpoint, remarkably effective strategy despite widespread reader fatigue and declining trust in digital media outlets.

The consequences of this practice extend beyond individual reader frustration. At the level of the media ecosystem, clickbait produces a race-to-the-bottom dynamic: outlets that resist sensationalist framing risk losing traffic — and consequently revenue — to competitors willing to exploit curiosity gaps more aggressively. At the level of public discourse, clickbait contributes to the broader crisis of trust in journalism, as readers who feel repeatedly misled by exaggerated headlines become more likely to disengage from legitimate news sources altogether or, conversely, become more susceptible to genuine misinformation once critical evaluation of headlines is abandoned as a coping strategy. It is for these reasons that automatic clickbait detection and mitigation has become an active area of research within Natural Language Processing (NLP), situated at the intersection of computational linguistics, information retrieval, and media studies.

Crucially, the problem of clickbait is best understood as having a **dual nature**. The first dimension — detection — asks a binary or graded question: *is this headline clickbait, and to what degree?* The second, comparatively under-explored dimension — which this work refers to as *resolution* — asks a generative question: *what is the actual answer that the headline withholds from the reader?* A system capable only of detection can flag or filter clickbait content, but it cannot restore to the reader the informational value that the headline promised. A system capable of resolution, by contrast, directly addresses the reader's original information need, either by extracting the relevant passage from the article body or by generating an abstractive summary that closes the curiosity gap without requiring the reader to navigate through filler content, advertising interstitials, and narrative misdirection. This distinction — already central to the NoticIA framework — is adopted as the organising principle of the present work: the object of study is not clickbait detection alone, but the joint task of clickbait characterization **and** answer resolution.

Figure 1.1

«Папочку» решили задобрить. НАТО опять пытается подольститься к Трампу

В ответ на требования Дональда Трампа передать США Гренландию НАТО запускает новую миссию на острове и в окружающих водах — «Арктический часовой». [...] Цели миссии также прописаны совсем невнятно: «Миссия станет частью усилий Североатлантического альянса по дальнейшему укреплению нашего сдерживания и обороны в [Арктическом] регионе [...]».

Hype, Teaser · Severity: 2 (Средняя)

Резюме (answer_summary): НАТО запускает добровольную миссию в Гренландии, цель — увеличить количество войск в регионе.

Бузова прокляла? Футболист Дмитрий Тарасов задолжал налоговой сотни тысяч рублей

Пока в интернете продолжают обсуждать недавнее интервью певицы Ольги Бузовой [...] у футболиста появилась очередная проблема. По информации Telegram-канала Shot, налоговая служба выставила Тарасову счёт ещё 28 декабря 2025 года. [...] к февралю 2026 года долг вырос до 279 000 рублей. [...] Действительно ли Бузова прокляла экс-избранника — неизвестно.

Hype, Teaser, Celeb, Question · Severity: 3 (Высокая)

Резюме (answer_summary): Дмитрий Тарасов задолжал ФНС 279 000 рублей.

Рютте назвал самую большую и сильную страну в мире

США являются «самой большой и сильной» страной во всём мире. С таким утверждением выступил генеральный секретарь Североатлантического альянса Марк Рютте [...] «США — самая большая и сильная страна в мире (...). Им нужно брать на себя ответственность за весь мир», — ответил генсек НАТО на вопрос журналистов [...].

Teaser, Vague · Severity: 2 (Средняя)

Резюме (answer_summary): США.

Figure 1.1: Examples of Russian-language clickbait headlines from the corpus constructed in this work, together with their human-annotated resolving summaries, illustrating the “curiosity gap” structure described in Section 1.1.

1.2 Problem Statement

Despite the growing body of NLP research on clickbait, the available resources are markedly skewed towards a small number of languages. Clickbait *detection* corpora exist for English (the Webis Clickbait Corpus 2017, comprising 38,517 annotated social media posts) and, more recently, for Spanish (the ES_Headline_Contradiction dataset). Clickbait *resolution* — the task of pairing a clickbait headline and its article body with a human-written answer — is an even scarcer resource. To the best of the author’s knowledge, at the time of writing, Noticia (García-Ferrero & Altuna, 2024) remains the only publicly available dataset offering headline–body–summary triplets explicitly designed to train and evaluate generative resolution systems, and it is constructed exclusively for Spanish.

This asymmetry constitutes a significant gap for Russian, a language spoken natively by over 150 million people and representing one of the largest non-English, non-Spanish digital news ecosystems globally. Three considerations motivate treating this gap as a genuine research problem rather than a matter of simple translation or transfer:

- **Linguistic specificity.** Russian morphology (rich case inflection, flexible word order, aspectual verb pairs) interacts with clickbait rhetorical devices — rhetorical questions, vague demonstrative pronouns (“это”, “такое”), truncated comparatives — in ways that are not necessarily homologous to their Spanish or English equivalents. A detection or resolution system trained on Spanish or English data, or a naively

machine-translated version of Noticia, cannot be assumed to transfer these rhetorical patterns without loss of fidelity.

- **Media-ecosystem specificity.** The categories of clickbait that dominate a given news ecosystem (celebrity-driven, health-related, politically framed, etc.) are shaped by local media culture, platform dynamics (e.g., the role of Telegram and Yandex Zen as distribution channels in the Russian-language digital sphere), and audience expectations, none of which are captured by existing non-Russian corpora. Empirical evidence from the neighbouring Belarusian media landscape corroborates this ecosystem-specificity: Hradziushko (2025), analysing the websites, social-media accounts, and YouTube channels of six major national outlets, documents recurrent Russian-language clickbait templates (e.g. headlines opening with numerals or sums of money, negative/tragedy framing, appeals to action) alongside platform-specific visual markers on YouTube (thumbnail close-ups, upper-case text, exclamatory punctuation) that have no direct counterpart in the Spanish- or English-language taxonomies underlying prior resolution corpora. **Absence of an annotated Russian resolution benchmark.** Without a dedicated corpus, it is not possible to rigorously evaluate — nor to fine-tune — LLMs on their capacity to understand and resolve Russian-language clickbait, leaving a blind spot in the otherwise rapidly advancing multilingual capabilities of modern LLMs.

The problem addressed by this work is therefore twofold: (i) the absence of an annotated Russian-language clickbait resolution corpus with the granularity required to support both classification and generative tasks, and (ii) the absence of empirical evidence regarding how state-of-the-art LLMs — evaluated both in a zero/few-shot regime and after targeted fine-tuning — perform on this task for Russian.

1.3 Motivation

The present work adopts the general methodology pioneered by Noticia — the construction of a headline–body–summary triplet corpus through careful human annotation, followed by systematic evaluation of LLMs in zero/few-shot and fine-tuned settings — while introducing two substantive extensions tailored to the Russian-language context and to the specific research questions of this TFM.

Extension 1: a graded, multi-label characterisation of clickbait, beyond a binary label.

Rather than treating clickbait purely as a binary detection target (clickbait / not clickbait), this work develops a formal taxonomy of ten binary linguistic and rhetorical features — *Intrigue/Incompleteness*, *Overpromise*, *Emotion/Hype*, *VaguePointer*, *RhetoricalQuestion*, *FOMO*, *Distortion*, *BrokenPromise*, *Formatting* (typographic markers such as capitalisation and exclamation clusters), and *Celebrity* exploitation. Each feature is assigned an empirically calibrated weight, from which a continuous *clickbait score* is computed and subsequently discretised into four severity classes (NONE, MILD, STRONG, EXTREME). This taxonomy, together with its weighting scheme, is described in full in Section 2.2 and operationalised in Section 3.3. Its purpose is to allow the resulting corpus to support not only a coarse detection

task but also fine-grained classification and severity regression tasks, substantially increasing its downstream reusability.

Extension 2: a comparative experimental design combining in-context learning and cross-lingual fine-tuning. Whereas Noticia’s experimental section is primarily organised around zero-shot evaluation of a broad model pool followed by fine-tuning models from scratch on the target task, this work adopts a design intended to directly address the research question of *language transfer*: to what extent can a model originally fine-tuned for clickbait resolution in Spanish (ClickbaitFighter-7B) be adapted to Russian through targeted fine-tuning, and how does the resulting model compare against a diverse pool of general-purpose LLMs — ranging from a 1-billion parameter Russian-oriented model to 70+ billion parameter multilingual models — evaluated through few-shot prompting alone? This comparison is intended to shed light on a practically important question for low-resource-relative-to-English languages: is it more effective to invest in cross-lingual fine-tuning of an existing task-specific model, or to rely on the emergent multilingual capabilities of large general-purpose models through in-context learning?

It should be stated explicitly, as a matter of academic honesty and scope definition, that the annotation of the corpus presented in this work was carried out by a single annotator (the author). This is a deliberate scope decision dictated by the resource constraints of a TFM, and it departs from the double-annotation protocol used in Noticia (which employed a second annotator on a validation subset to compute inter-annotator agreement). The implications of this design choice for corpus reliability are discussed critically in Section 3.3, and its mitigation through future multi-annotator validation is proposed as a direction for future work in Section 6.2. This limitation notwithstanding, single-annotator corpus construction remains a common and accepted practice in resource-constrained NLP research, provided that annotation guidelines are made explicit, systematic, and — as is the case in this work — grounded in an auditable, formalised scoring methodology rather than in unstructured subjective judgement.

Finally, it must be emphasised that the construction of the corpus itself — rather than the modelling experiments that follow — constitutes the methodological core of this TFM. Approximately 90% of the total effort invested in this project was dedicated to the annotation process: locating suitable clickbait articles, formulating the implicit question posed by each headline, identifying and extracting the passage of the article body that resolves that question, and assigning the corresponding taxonomy labels and severity scores. This emphasis is not incidental. As is well established in the NLP literature and explicitly demonstrated by García-Ferrero and Altuna (2024) — who report that ROUGE performance on their fine-tuned models had not yet saturated at 750 training samples — the quality, consistency, and granularity of a resolution corpus directly bound the achievable performance of any model trained or evaluated on it, regardless of model scale or prompting strategy. A central design goal of this work, therefore, was to produce a Russian-language corpus that is not only usable for the specific experiments reported in Chapters 4 and 5, but that is *structurally scalable*: modular enough in its schema (raw full-text fields, balanced and severity-scored variants, and model-ready reduced-column exports) to be extended with additional annotated examples,

additional annotators, or integrated into other Russian-language NLP resources beyond the scope of this TFM.

1.4 Aims and Objectives

The **general aim** of this work is to design, construct, and validate a Russian-language corpus for clickbait characterisation and resolution, and to empirically evaluate the extent to which current-generation Large Language Models — under in-context learning and under targeted cross-lingual fine-tuning — are capable of performing the resolution task on this corpus.

This general aim is decomposed into the following **specific objectives**:

- To design and implement an automated data-collection pipeline (web crawling) capable of retrieving candidate clickbait news articles from Russian-language digital media sources, together with their full text and associated metadata.
- To design a structured, extensible schema for organising the collected articles into a unified corpus, distinguishing clickbait and non-clickbait samples and preserving the metadata required for subsequent annotation and analysis.
- To develop and apply a formalised, weighted taxonomy of ten clickbait linguistic features, enabling the assignment, to each annotated headline, of (a) a set of categorical tags, (b) a continuous clickbait score, and
- a discrete severity class, in addition to a human-written answer summary that resolves the headline’s implicit question.
- To manually annotate a corpus of Russian-language clickbait articles following the taxonomy and guidelines defined in Objective 3, producing headline–body–answer triplets analogous to those of the NoticIA dataset.
- To pre-process and prepare the annotated corpus for consumption by language models, including the construction of balanced and full corpus variants and the reduction of the schema to the minimal fields required for prompting and fine-tuning.
- To evaluate, in a few-shot in-context learning regime, the resolution capability of four language models of varying scale and origin — namely Qwen2.5-72B-Instruct, Qwen3-VL-32B-Instruct, Qwen2.5-7B-Instruct, and Llama-3.1-70B-Instruct, all accessed through the OpenRouter API — using automatic metrics for generation quality.
- To fine-tune ClickbaitFighter-7B — a model originally trained for clickbait resolution in Spanish — on the Russian-language annotated corpus, in order to assess the effectiveness of cross-lingual adaptation of a task-specific model relative to few-shot prompting of general-purpose LLMs.
- To establish an evaluation protocol based on BLEU, ROUGE-L, and average answer length, enabling a systematic comparative analysis of the five resulting systems (four

few-shot models and one fine-tuned model) against the human-annotated gold-standard answers.

- To critically discuss the results obtained, the limitations inherent to the single-annotator corpus design, and concrete directions for extending the dataset and improving the modelling approaches in future work.

1.5 Structure of the Document

The remainder of this document is organised as follows. **Chapter 2** reviews the state of the art in clickbait detection and resolution, situating this work with respect to the NoticIA dataset, the ClickbaitFighter model family, and the broader SEPLN publication, and presents the theoretical taxonomy of clickbait categories that underpins the annotation methodology of this work. It further introduces the theoretical foundations of the modelling approaches employed later, namely the Transformer architecture, few-shot in-context learning, and fine-tuning. **Chapter 3** describes the methodology followed to construct the corpus, in four stages: data collection through web crawling, corpus structuring, manual annotation, and pre-processing for model consumption. **Chapter 4** presents the experimental setup, covering both the few-shot evaluation of four LLMs and the fine-tuning of ClickbaitFighter-7B on the Russian corpus. **Chapter 5** reports and analyses the results of these experiments according to the BLEU, ROUGE-L, and average-length metrics defined in Chapter 4. Finally, **Chapter 6** summarises the conclusions of the work with respect to the objectives stated above and outlines directions for future work, including the extension of the corpus through multi-annotator validation and further model improvements.

2. Estado del Arte / Marco Teórico

2.1 Foundational Projects: NoticIA, ClickbaitFighter, and the SEPLN Publication

2.1.1 Prior work on clickbait detection

Automatic clickbait detection has been studied for over a decade. A watershed moment for the field was the Clickbait Challenge organised by Potthast et al. (2018), which brought together twelve competing teams and resulted in the release of the Webis Clickbait Corpus 2017, a collection of 38,517 annotated social-media posts that has since served as a reference benchmark for numerous detection systems, including the work of Zheng, Yu, and Wu (2021). Detection-oriented research has continued to expand in recent years, with contributions such as those of Pujahari and Sisodia (2021), Liu et al. (2021), and Wang, Maslim, and Liu (2023) applying increasingly sophisticated classification architectures to the binary or multi-class detection task (see References for full bibliographic details).

However, the present work is concerned less with the binary detection task than with what has come to be known as *clickbait spoiling* or *resolution*: the generation or extraction of the information that the clickbait headline withholds. The PAN Clickbait Challenge at SemEval

2023 (Fröbe et al., 2023) formalised this task in two complementary sub-problems — classifying the structural type of the spoiler required (phrase, passage, or multi-part) and generating the spoiler itself — evaluating twenty-three participating systems on the Webis Clickbait Spoiling Corpus 2022, a corpus of 5,000 headline–news–spoiler triplets collected from social media. A related line of work frames the task explicitly as question answering: Heiervang (2022) proposes a title-answering method in which fine-tuned LLMs extract an abstractive summary from the article body to answer the implicit question posed by the headline, while Kurenkov et al. (2022) fine-tune a RoBERTa model to produce both extractive and abstractive answers, and Intan Maharani, Purwarianti, and Aji (2023) instead formulate the task as a single- or multi-span information-extraction problem. Closer to the present work’s linguistic focus, Sepúlveda-Torres, Bonet-Jover, and Saquete (2023) study misleading headlines in Spanish news through the ES_Headline_Contradiction dataset — 18,542 headline–body pairs annotated for contradiction between headline and content — and fine-tune BERT-based models to detect and classify these contradictions with high accuracy.

This body of work establishes two important precedents that directly inform the design of the present TFM: first, that clickbait resolution can be productively framed as a constrained question-answering or summarisation task rather than as open-ended generation; and second, that the quality of such systems is fundamentally bound by the availability of a high-granularity annotated corpus in the target language — a resource that, prior to this work, did not exist for Russian.

2.1.2 The Noticia dataset

The single most direct antecedent of this TFM is Noticia (García-Ferrero & Altuna, 2024), a dataset of 850 Spanish news articles, each pairing a prominent clickbait headline with its full article body and a high-quality, single-sentence, human-written summary that resolves the headline’s implicit question. Noticia is, to date, the only publicly available dataset offering complete headline–body–*ultrasummary* triplets explicitly designed to benchmark the information-retrieval and text-comprehension capabilities of LLMs under this task formulation.

Two aspects of Noticia’s construction are of particular methodological relevance to this work. First, its corpus-collection strategy deliberately avoided massive undirected web crawling, instead exploiting two readily-available sources of pre-identified clickbait headlines: social media accounts that manually curate and summarise clickbait articles, and a public web application built from an early version of the summarisation model itself, which was used to solicit additional user-submitted examples. Second, its annotation protocol was carried out by PhD students with native-level competence in the target language, instructed first to identify the implicit question posed by the headline and subsequently to locate the exact passage of the article body that answers it — a two-step *question identification, then answer localisation* procedure that is directly adopted, and extended with a formal severity-classification layer, in the annotation methodology described in Chapter 3 of this work.

Noticia further reports a systematic zero-shot evaluation of over thirty LLMs, ranging from 1.1 to 72 billion parameters, using ROUGE-L as the primary metric, alongside the fine-tuning

of three task-specific models — ClickbaitFighter-2B, -7B, and -10B — described in the following subsection. Two empirical findings from this evaluation are particularly consequential for the design of the present work. First, model scale alone is a poor predictor of resolution quality: models trained on comparatively small but high-quality instruction datasets (e.g. OpenChat-3.5-0106, at 7 billion parameters) markedly outperformed substantially larger general-purpose models such as Llama-2-70b-chat-hf. Second, and more importantly for the motivation of this TFM’s corpus-quality emphasis (Section 1.3), a controlled experiment varying the number of fine-tuning samples showed that ROUGE performance had not yet plateaued at 750 training examples — that is, the ceiling imposed by dataset size, rather than by model capacity, remained the binding constraint on system performance. This finding directly underpins the decision, in the present work, to prioritise annotation depth and corpus scalability over sheer model-experiment breadth.

Dimension	Noticia (García-Ferrero & Altuna, 2024)	This work
Language	Spanish	Russian
Number of articles (total)	850	1,296 (600 clickbait + 696 non-clickbait)
Number of clickbait examples	850 (100%)	600 (46.3%)
Number of annotators	2 (main + independent validation annotator on a subset)	1 (single annotator; see Section 3.3.2)
Severity / tag annotation	Not provided (binary clickbait status only, no graded taxonomy)	10-feature weighted taxonomy + continuous severity score + 4-level severity class (Section 2.2)
Non-clickbait counterpart	Not provided	Provided (594 leakage-audited non-clickbait rows, Section 3.2.2)
Provenance / collection method	Pre-curated clickbait source (X/Twitter account) + web-app submissions	Comprehensive web crawling of 4 outlets, undifferentiated news stream (Section 3.1.1)
License / availability	Publicly released (Hugging Face: Iker/Noticia, somosnlp/Noticia-it)	Not yet publicly released; release planned upon submission of this TFM

Table 2.1: Summary comparison between the Noticia dataset (García-Ferrero & Altuna, 2024) and the Russian-language corpus constructed in this work.

2.1.3 The ClickbaitFighter model family and the SEPLN publication

Noticia was published as a peer-reviewed article associated with the SEPLN (Sociedad Española para el Procesamiento del Lenguaje Natural) venue, and its authors additionally released the ClickbaitFighter family of task-specific fine-tuned models, in three sizes:

ClickbaitFighter-2B (based on Gemma-2B-IT), ClickbaitFighter-7B (based on openchat-3.5-0106), and ClickbaitFighter-10B (based on Nous-Hermes-2-SOLAR-10.7B). All three models were fully fine-tuned — that is, without parameter-efficient adapters — on the 700-example Noticia training split, using bfloat16 precision and DeepSpeed Zero-3 (Ren et al., 2021) to distribute optimisation across four NVIDIA A100 80GB GPUs, with the standard Next-Token-Prediction loss computed exclusively over the summary tokens (masking the loss on the article-body tokens) so as to prevent the — typically much longer — input article from overwhelming the training signal associated with the target summary.

The reported results establish a clear performance hierarchy: the human inter-annotator baseline achieves a ROUGE-L score of 55.98 with an average summary length of 11.03 words; ClickbaitFighter-10B and -7B achieve 52.01 and 49.81 respectively, approaching human-level quality; and even the smallest variant, ClickbaitFighter-2B, at 36.26, outperforms every evaluated zero-shot general-purpose model, including GPT-4o and Llama-3-70B-Instruct. This result — a small, task-specific fine-tuned model outperforming substantially larger general-purpose models used zero-shot — constitutes one of the central empirical motivations for the comparative experimental design adopted in Chapter 4 of this work, in which ClickbaitFighter-7B is fine-tuned on Russian-language data and directly compared against four general-purpose LLMs used in a few-shot regime. The present work is thus best understood not as a replication of Noticia, but as a targeted test of whether the task-specific advantage documented for Spanish transfers to Russian through cross-lingual fine-tuning of the same base architecture.

2.2 Theoretical Classification of Clickbait

While Noticia’s annotation protocol is centred on the extraction of a resolving answer, it does not provide a formalised, graded taxonomy of *why* a given headline qualifies as clickbait, nor a mechanism for expressing *degrees* of clickbait severity. This gap motivated the development, within this TFM, of an original ten-feature taxonomy together with a weighted scoring formula, described in this section. This taxonomy was designed specifically to support the annotation process described in Chapter 3 and to allow the resulting corpus to serve, beyond its resolution-generation purpose, as a multi-label classification and severity-regression benchmark.

2.2.1 The ten-feature taxonomy

Each headline in the corpus is characterised along ten binary linguistic and rhetorical features, denoted $p_1, \dots, p_{10} \in \{0,1\}$, each corresponding to a distinct, independently-attestable manipulative device. The features are grouped into two classes according to their manipulative intensity: *strong* features, which directly and aggressively exploit the reader’s curiosity or emotional state ($p_1, p_2, p_3, p_6, p_7, p_8$), and *weaker*, more formal or auxiliary features, which intensify but do not by themselves constitute clickbait (p_4, p_5, p_9, p_{10}). The ten features, together with their single-word tag equivalents used in the `clickbait_type` field of the corpus, are defined as follows:

- **Intrigue / Incompleteness (TEASER, p1).** The headline deliberately creates an information gap by withholding the core fact it refers to, compelling the reader to click in order to resolve the gap.
- **Overpromise (OVERPROMISE, p2).** The headline promises a radical, immediate, or universally applicable outcome without qualification — e.g. guarantees, “forever” framing, or “miracle solution” framing — without stating the conditions or limitations under which the outcome holds.
- **Emotion / Shock (HYPE, p3).** The headline substitutes neutral, factual description with emotionally charged vocabulary — shock, horror, scandal — engineered to provoke an affective reaction rather than to inform.
- **Vague Pointer (VAGUE, p4).** The headline relies on demonstrative pronouns or deictic expressions (“this”, “such a”, “this is why...”) in place of the concrete referent, manufacturing curiosity through grammatical rather than semantic incompleteness.
- **Rhetorical Question (QUESTION, p5).** The headline is phrased as a question whose answer is deliberately withheld and functions purely as bait, as opposed to a genuine informational question answered within the headline itself.
- **FOMO / Urgency (FOMO, p6).** The headline invokes scarcity, exclusivity, or time pressure (“only today”, “before it’s deleted”) to accelerate the reader’s decision to click.
- **Distortion / Exaggeration (DISTORT, p7).** The headline disproportionately inflates the scale or significance of the reported event relative to what the article body actually supports.
- **Broken Promise (BROKEN, p8).** The most severe feature: the headline makes an explicit promise that the article body does not, or only partially, fulfils — a direct content mismatch between headline and substance.
- **Loud Formatting (LOUD, p9).** Purely typographic intensifiers: excessive capitalisation, clustered punctuation (“?!”, ellipses), or an informal second-person address (“you won’t believe...”) used to amplify the perceived urgency of the headline.
- **Celebrity / Brand Exploitation (CELEB, p10).** The headline exploits the recognisability of a public figure’s name or a brand without that figure or brand playing a substantively informative role in the article content.

Each feature is annotated as a binary indicator per headline, and a headline may simultaneously exhibit any subset of these ten features; the annotation methodology by which each feature is scored for a given article is described in Section 3.3.

Figure 2.1: The ten-feature clickbait taxonomy developed for this work, grouped by manipulative intensity (strong vs. weaker/auxiliary features) and mapped onto the three derived corpus representations described in Section 2.2.4.

2.2.2 Weighting scheme and the clickbait severity formula

To move from a purely categorical (tag-based) representation to a continuous, comparable severity measure, each feature p_i is assigned an empirically motivated weight w_i , reflecting its relative manipulative intensity as established in the qualitative analysis of Section 2.2.1. The weights adopted in this work are:

$$w_1 = 2.0 \text{ (Intrigue)}$$

$$w_2 = 2.0 \text{ (Overpromise)}$$

$$w_3 = 1.5 \text{ (Emotion)}$$

$$w_4 = 1.0 \text{ (VaguePointer)}$$

$$w_5 = 1.0 \text{ (RhetQuestion)}$$

$$w_6 = 1.5 \text{ (FOMO)}$$

$$w_7 = 2.0 \text{ (Distortion)}$$

$$w_8 = 2.5 \text{ (BrokenPromise)}$$

$$w_9 = 0.5 \text{ (Formatting)}$$

$$w_{10} = 0.5 \text{ (Celebrity)}$$

The weight of 2.5 assigned to *BrokenPromise* is the highest in the scheme, reflecting the position — adopted throughout this work — that a direct mismatch between what a headline promises and what the article actually delivers constitutes the most consequential form of reader deception, exceeding purely emotional or urgency-based manipulation in severity. By contrast, *Formatting* and *Celebrity* receive the lowest weight (0.5), as both are treated as amplifying rather than constitutive features: capital letters or a recognisable name alone do not make a headline clickbait in the absence of an underlying informational gap.

Given a headline's binary feature vector $p = p_1, \dots, p_{10}$, the raw clickbait score is computed as the weighted sum:

$$\text{Score} = \sum_{i=1}^{10} w_i \cdot p_i$$

which is then normalised to the 0,1 interval by dividing by the theoretical maximum attainable score — the sum of all ten weights, $\sum_{i=1}^{10} w_i = 14.0$:

$$\text{Score}_{\text{norm}} = \text{Score} / 14.0$$

2.2.3 Empirical recalibration of severity thresholds

An initial, purely theoretical discretisation of $\text{Score}_{\text{norm}}$ into four ordinal severity classes was proposed prior to annotation, using evenly-motivated thresholds: NONE (<0.2), MILD ($0.2-0.4$), STRONG ($0.4-0.7$), and EXTREME (>0.7). However, once applied to the actually annotated corpus, this theoretical discretisation proved empirically inadequate. A

distributional analysis of the number of tags assigned per headline (Table 2.2) revealed that annotators — following the guidelines of Section 3.3 — essentially never assigned all ten features simultaneously to a single headline: the dominant pattern in the corpus is a combination of two to three tags, jointly accounting for approximately 78% of all annotated clickbait samples, with four- and five-tag combinations occurring in only 5.7% and 0.7% of cases respectively.

Number of tags	Count	Percentage	Interpretation
1	93	15.5%	Minimal clickbait — a single rhetorical device present
2	290	48.3%	Dominant pattern — two reinforcing devices (e.g. Hype + Teaser)
3	179	29.8%	Elevated manipulation — three co-occurring devices
4	34	5.7%	Aggressive, multi-device clickbait (uncommon)
5	4	0.7%	Extreme / maximal combination (rarest pattern)
Total	600	100.0%	2–3 tags jointly account for 78.1% of the corpus

Table 2.2: Empirical distribution of tag co-occurrence per headline, showing that 2–3 tags constitute the dominant clickbait pattern (78.1% of the annotated corpus, N=600).

Because dividing the raw score by the absolute theoretical maximum of 14.0 implicitly assumes that headlines exhibiting all ten features are a realistic occurrence, applying it directly compressed the vast majority of the corpus (459 of the 600 annotated clickbait rows) into the lower NONE/MILD range, leaving the STRONG and EXTREME classes almost empty. Such class collapse is undesirable both for descriptive purposes — it fails to discriminate between headlines that a human reader would clearly judge as markedly different in manipulateness — and for any downstream supervised model (e.g. a BERT-based severity classifier) trained on these labels, which would receive an overwhelmingly skewed and largely uninformative target distribution.

To correct this “flattening” effect while preserving the underlying linear weighting logic of Section 2.2.2, the severity thresholds were recalibrated against the empirically observed score distribution of the corpus, rather than against the abstract theoretical maximum, yielding the following four-class scheme actually applied to the corpus (implemented in 3_Анализ_датасета_TFM.ipynb):

- 1. Low severity (formatting / weak):** $\text{Score}_{\text{norm}} \leq 0.15$
- 2. Medium severity (moderately manipulative):** $0.15 < \text{Score}_{\text{norm}} \leq 0.30$
- 3. High severity (pronounced clickbait):** $0.30 < \text{Score}_{\text{norm}} \leq 0.50$
- 4. Critical severity (aggressive clickbait):** $\text{Score}_{\text{norm}} > 0.50$

This recalibration episode is reported here not merely as an implementation detail but as a methodologically instructive result in its own right: it demonstrates that a severity metric

designed on purely a priori theoretical grounds must be validated — and, where necessary, recalibrated — against the empirical distribution of the annotated data before being adopted as a supervised learning target. The resulting `severity_score` (continuous) and `severity_class` (four-level ordinal) fields are retained as corpus columns and constitute one of the extensions that distinguish this corpus from NoticIA, as discussed in Section 1.3.

Class	Label	Count	Percentage
1	Low (formatting/weak)	92	15.3%
2	Medium (moderately manipulative)	364	60.7%
3	High (pronounced clickbait)	139	23.2%
4	Critical (aggressive)	5	0.8%
Total		600	100.0%

Table 2.3: Distribution of severity classes in the annotated corpus following empirical threshold recalibration (N=600).

2.2.4 Structured representation

For each annotated headline, the corpus stores three complementary layers of clickbait characterisation, jointly derived from the taxonomy described above: (i) a set of categorical tags corresponding to the features attested in the headline (e.g. TEASER, OVERPROMISE, HYPE); (ii) a continuous `clickbait_score` (equivalently, `severity_score`) in 0,1; and (iii) a discrete `severity_class` label. This triple representation is intended to maximise the reusability of the corpus: the categorical layer supports multi-label classification and qualitative linguistic analysis, the continuous layer supports regression-style modelling, and the discrete layer supports conventional ordinal or multi-class classification, without requiring re-annotation for any of these three downstream uses.

2.3 Modern Approaches to Automation: Transformers, LLMs, Few-shot Learning, and Fine-tuning

2.3.1 The Transformer architecture

The modelling approaches evaluated in this work — both the general-purpose LLMs used under few-shot prompting and the ClickbaitFighter model fine-tuned in Chapter 4 — are built upon the Transformer architecture introduced by Vaswani et al. (2017). Unlike recurrent architectures, the Transformer relies entirely on the self-attention mechanism to model dependencies between tokens regardless of their distance in the sequence, which affords both substantially improved parallelisation during training and a demonstrated capacity to capture long-range discourse dependencies — a property of direct relevance to the clickbait-resolution task, in which the answer to a headline is frequently located many sentences away, at the end of a lengthy article body, separated from the headline by extensive filler content. Autoregressive, decoder-only Transformer variants, trained on the next-token-prediction objective over massive text corpora, constitute the architectural basis of all models evaluated

in this work, from the 1-billion-parameter end of the scale up to 70-billion-parameter general-purpose models.

2.3.2 Large Language Models and emergent capabilities

Scaling decoder-only Transformers to tens or hundreds of billions of parameters, combined with instruction-tuning and human-feedback alignment procedures, has given rise to what is now generically termed Large Language Models (LLMs). Of particular relevance to this work is the empirically documented phenomenon of *emergent multilingual competence*: models trained predominantly on English-majority corpora nonetheless exhibit non-trivial competence in typologically and orthographically distant languages, including Russian, despite the comparatively small proportion of Russian-language tokens in most public pre-training corpora. The experimental design of Chapter 4 — evaluating four general-purpose LLMs of markedly different scale and provenance on a Russian-language task via few-shot prompting — is directly motivated by the need to empirically characterise the extent and reliability of this emergent multilingual competence for the specific, comparatively demanding task of clickbait resolution, which, as argued in Section 1.1, requires not merely surface-level language competence but discourse-level comprehension of an entire article body.

2.3.3 Few-shot in-context learning

Few-shot in-context learning, as characterised by Brown et al. (2020) in the context of GPT-3, refers to the capacity of sufficiently large language models to adapt to a novel task at inference time, without any gradient-based parameter update, purely by conditioning on a small number of input–output demonstration pairs prepended to the prompt. In the present work, few-shot prompting is operationalised, for each of the four evaluated models, through a fixed system instruction that specifies the task constraints — namely, that the model must produce a single, concise, factual sentence in Russian that resolves the informational gap posed by the clickbait headline, without introductory phrases such as “Answer:” — followed by one worked example (headline, article excerpt, and target answer) and the test instance to be resolved. This design choice — a single demonstration example, rather than the multiple-shot prompts more commonly reported in the few-shot learning literature — reflects a deliberate trade-off, discussed further in Section 4.2, between demonstration coverage and the practical context-length and cost constraints imposed by processing the full, often lengthy, article bodies characteristic of clickbait content.

The principal appeal of the few-shot approach, and the reason it is retained as one of the two experimental conditions compared in this work, is its near-zero adaptation cost: it requires no annotated training split, no gradient computation, and no task-specific model storage, making it the default strategy of choice whenever a sufficiently capable general-purpose model is available off-the-shelf. Its principal limitation, conversely, is that model behaviour under few-shot prompting can be comparatively brittle and sensitive to prompt phrasing, and — critically for this work’s research question — is entirely contingent on whatever multilingual and task-relevant competence the base model acquired during pre-training, without any

opportunity to correct systematic errors through exposure to task-specific, language-specific supervision.

2.3.4 Fine-tuning

Fine-tuning, by contrast, involves updating a pre-trained model’s parameters — in this work, all parameters, following the full-parameter fine-tuning protocol of the original ClickbaitFighter models described in Section 2.1.3, rather than a parameter-efficient adaptation method such as LoRA (Hu et al., 2022) — on a task-specific and, in this case, language-specific annotated corpus, using a supervised training objective. As detailed in Section 2.1.3, the original ClickbaitFighter models were trained with the standard next-token-prediction loss computed exclusively over the target-summary tokens, a design choice retained in the cross-lingual fine-tuning methodology described in Chapter 4 of this work, in order to preserve comparability between the Spanish-language training recipe reported by García-Ferrero and Altuna (2024) and its Russian-language adaptation performed here.

The central research question motivating the inclusion of fine-tuning as an experimental condition in this work — rather than relying on few-shot prompting alone — follows directly from the empirical result discussed in Section 2.1.3: that a comparatively small, task-specific fine-tuned model (ClickbaitFighter-7B, 7 billion parameters) outperformed substantially larger general-purpose LLMs used zero-shot on the same task for Spanish. This work tests whether an analogous advantage can be obtained for Russian not by training a task-specific model from scratch, but through *cross-lingual fine-tuning*: adapting a model whose parameters already encode task-specific competence for clickbait resolution in one language (Spanish) to a second, typologically distant language (Russian), using a comparatively small annotated corpus. This cross-lingual transfer setting — as opposed to either monolingual fine-tuning or zero-shot cross-lingual transfer without any target-language fine-tuning — occupies a middle ground that, to the best of the author’s knowledge, has not been previously documented for the specific case of Spanish-to-Russian clickbait resolution, and constitutes one of the principal empirical contributions of this TFM. The full experimental protocol, including hyperparameter configuration and adaptation-specific considerations (e.g. tokenizer coverage of Cyrillic script, catastrophic forgetting of the Spanish source task), is described in Section 4.3.

2.3.5 Rationale for the comparative design

The joint evaluation of few-shot prompting and fine-tuning, rather than the adoption of a single modelling paradigm, is motivated by the fact that these two approaches occupy genuinely distinct points in the cost–performance–adaptability trade-off space relevant to any practitioner seeking to deploy a clickbait-resolution system for a new language. Few-shot prompting of an existing general-purpose LLM requires no annotated corpus beyond a handful of demonstration examples and no computational investment in training, but offers no guarantee of task-specific reliability and remains dependent on whatever proprietary or open-weight model happens to be available and sufficiently multilingual. Fine-tuning, conversely, requires an annotated corpus of the kind constructed in Chapter 3 of this work and non-trivial computational resources, but offers the possibility of task-specific and language-specific

specialisation beyond what any general-purpose model can be expected to provide out-of-the-box. By evaluating both paradigms on the same corpus and under the same evaluation protocol (Chapter 5), this work aims to provide an empirically grounded answer, specific to the Russian clickbait-resolution task, to the practically consequential question of which paradigm offers the better return on the investment required to construct a dedicated annotated corpus.

3. Metodología y Creación del Corpus

As stated in Chapter 1, the construction of the corpus constitutes the methodological centre of gravity of this TFM: approximately 90% of the total time invested in the project was dedicated to data collection, structuring, and — above all — manual annotation, substantially outweighing the time subsequently spent on the modelling experiments reported in Chapter 4. This chapter documents that process in the four stages announced in Section 1.5: (i) automated data collection through web crawling, (ii) corpus structuring and consolidation, (iii) manual annotation, and (iv) pre-processing of the annotated corpus into model-consumable variants.

3.1 Stage 1: Data Collection

3.1.1 Design rationale: continuous news-stream sampling versus curated clickbait sources

A first, consequential methodological decision distinguishes the corpus construction strategy of this work from that of Noticia. As described in Section 2.1.2, García-Ferrero and Altuna (2024) deliberately avoided undirected web crawling and instead built 725 of their 850 articles (85%) from a single, pre-curated source: a social media account that manually identifies and republishes already-recognised clickbait headlines. This strategy is efficient — it guarantees a high density of genuine clickbait examples per unit of collection effort — but it also means that the resulting corpus is not representative of the *undifferentiated* stream of headlines an ordinary reader encounters when browsing a news outlet’s homepage or RSS feed, where clickbait and non-clickbait headlines are interleaved and the reader must exercise their own, unaided judgement to tell them apart.

The present work adopts the opposite strategy. Rather than sourcing pre-identified clickbait examples, four Russian-language news outlets — **eg.ru**, **aif.ru**, **lenta.ru**, and **mail.ru** — were crawled directly and comprehensively over their available publication history, without any pre-filtering for clickbait content, so as to reproduce the undifferentiated stream of headlines that characterises real-world, “in-the-wild” news consumption. This design choice was motivated by the explicit goal, alongside the resolution task inherited from Noticia, of also supporting a *realistic detection* task: training and evaluating models to distinguish clickbait from ordinary reporting under conditions that resemble the rapid, low-effort headline-scanning behaviour of an actual reader, rather than under the comparatively easier condition of discriminating between a pre-filtered clickbait sample and an unrelated control set. The

consequences of this design decision for the corpus’s class-balancing strategy are detailed in Section 3.4.

A practical limitation of this approach, inherent to the reliance on publicly accessible RSS feeds and article listings rather than a paid archival API, is that the target outlets do not retain openly accessible article listings indefinitely, and the effective temporal depth reachable through each outlet’s public infrastructure varies considerably by source and by collection method (Table 3.1). For the two outlets collected purely through RSS feeds, aif.ru and lenta.ru, the retrievable window at the time of crawling was narrow: their feeds exposed only the most recent seven to eight days of publications, reflecting the fact that RSS feeds are designed to surface the latest output rather than to serve as a historical archive. mail.ru’s feed exposed a substantially wider window of approximately four months (126 days). eg.ru, collected through a Scrapy crawl-spider that follows internal “related articles” links rather than being restricted to a single feed (Section 3.1.2), reached further back still: the large majority of its articles (320 of 384, 83%) fall within a seven-month window, but a residual 64 articles (17%) were reached via archival cross-links dating back to 2020, an artefact of the link-following collection strategy rather than a deliberately sampled historical period. Despite this source-level heterogeneity, the overall design rationale is unaffected: none of the four sources was collected by deliberately targeting a chosen historical window, and the resulting corpus in every case reflects whatever publication history each outlet’s public infrastructure happened to expose at the time of crawling, consistent with the “continuous stream” sampling rationale described above.

Source	Min. date	Max. date	Days covered	N articles	Collection method
aif.ru	2026-02-06	2026-02-12	7	472	RSS feed
eg.ru	2025-07-25 ¹	2026-02-12	203 ¹	384	Scrapy crawl-spider
lenta.ru	2026-02-05	2026-02-12	8	1,271	RSS feed
mail.ru	2025-10-16	2026-02-18	126	300	RSS feed
Overall (raw)	2020-01-31 ¹	2026-02-18	—	2,427	4 sources combined

¹ For eg.ru, 320 of 384 articles (83%) fall within a coherent 203-day window (2025-07-25 to 2026-02-12); the remaining 64 articles (17%) were reached via archival “related articles” cross-links followed by the Scrapy crawl-spider (Section 3.1.2) and date back to 2020, which is why the “Overall (raw)” row’s minimum date and the days-covered figure are dominated by this collection artefact rather than by a deliberately sampled historical period.

Table 3.1: Temporal coverage of the raw corpus per news outlet (N=2,427).

3.1.2 Crawler architecture per source

The data-collection pipeline, implemented in 1_recohida_de_los_datos_ TFM.ipynb, combines two complementary scraping strategies depending on the technical characteristics of each target outlet: a lightweight RSS-plus-BeautifulSoup approach for outlets exposing structured RSS feeds, and a Scrapy-based crawl-spider approach for outlets requiring deeper site navigation.

aif.ru and lenta.ru (RSS + BeautifulSoup). For these two outlets, the crawler first retrieves a fixed set of category-specific RSS feeds (for aif.ru: Политика, Общество, Экономика, Происшествия; for lenta.ru: a broader set spanning Главное, Россия, Экономика, Мир, Интернет и СМИ, Силовые структуры, and Наука и техника), parses each feed's entries with BeautifulSoup's XML parser to extract the article URL, headline, publication date, and source category, and subsequently dereferences each individual article URL to extract the full article body from the outlet-specific HTML container (e.g. `div.article_text p`, `.article__text p` for aif.ru; `.topic-body__content p`, `.topic-content p` for lenta.ru), concatenating all paragraph-level text nodes into a single text field. A custom desktop User-Agent header is set on all requests to reduce the likelihood of being served a degraded or bot-specific response, and a short request timeout is enforced per article to bound the total crawl duration.

eg.ru (Scrapy CrawlSpider). Because eg.ru does not expose a sufficiently granular RSS feed for the required categories, this source is instead crawled using a Scrapy-based spider seeded from five section-level entry points (showbusiness, politics, society, news, relax), which follows internal links matching the outlet's article URL pattern, extracts the headline and body text from each resulting article page, and discards short or malformed pages (through a minimum text-length filter of 200 characters) to reduce noise in the resulting corpus. A polite crawl-delay of 0.3 seconds between requests is configured to limit server load. Category labels harvested from this source use outlet-specific naming conventions (e.g. ЗВЕЗДЫ, РЕЗОНАНС), which are subsequently mapped onto the harmonised category scheme used across all four sources (see Section 3.2).

mail.ru. This source is likewise collected through a combination of feed-based link discovery and per-article HTML extraction, covering the categories Общество, Политика, Происшествия, СВО, and Экономика, contributing the smallest of the four per-source sub-corpora (300 unique articles after de-duplication).

Across all four sources, duplicate articles — identified by exact URL match — are removed at the point of collection, and publication dates are normalised to a single YYYY-MM-DD HH:MM format to support later chronological analysis. The four resulting per-source master files (`aif_thesis_master.csv`, `lenta_full_text.csv`, `eg_thesis_master.csv`, `mail_thesis_master.csv`) constitute the raw output of Stage 1.

3.1.3 Raw corpus yield

The four per-source crawls jointly yielded **2,427 unique articles** (aif.ru: 472; eg.ru: 384; lenta.ru: 1,271; mail.ru: 300), consolidated into the raw dataset `Data_raw.xlsx`. This raw yield already reflects the URL-level de-duplication performed independently within each source during Stage 1; cross-source de-duplication and category harmonisation are addressed in Stage 2, described in Section 3.2.

3.2 Stage 2: Corpus Structuring

3.2.1 Consolidation and category harmonisation

The four per-source master files are merged into a single working corpus in `2_Arreglamiento_de_los_datos.ipynb`. Because each outlet organises its content under a distinct, source-specific set of category labels (e.g. `eg.ru`'s `ЗВЕЗДЫ` versus `lenta.ru`'s native RSS category names), a category-mapping step harmonises these labels into a shared taxonomy (`Общество`, `Политика и Безопасность`, `Происшествия`, `Шоу-бизнес`, `Экономика`, and several source-specific residual categories for `lenta.ru`, such as `Мир`, `Наука и техника`, and `Силловые структуры`) applied consistently across all four sources. This step is a necessary precondition for the category-level leakage analysis described in Section 3.2.2, since that analysis is only meaningful if category labels carry comparable semantics across sources.

3.2.2 Diagnosing and mitigating source/category leakage

A central methodological contribution of this stage — and one of the clearest illustrations of the annotation-adjacent analytical effort this work invested in corpus quality — is a systematic diagnostic analysis of the joint distribution of source and category across the clickbait and non-clickbait subsets, carried out in `3_Анализ_датасета_TFM.ipynb` once the annotation described in Section 3.3 had been completed. This analysis directly addresses a risk well known in supervised NLP as *shortcut learning* or, in the terminology used during the project's own internal review, the “*lazy model*” problem: if clickbait and non-clickbait examples are systematically associated with different sources or topical categories, a classifier trained on the corpus may learn to exploit these superficial correlations — e.g. “if source is `eg.ru` and category is `Шоу-бизнес`, predict clickbait” — instead of learning the intended linguistic and rhetorical markers of clickbait described in Section 2.2.

The cross-tabulation of source $\times$ category for the clickbait subset (600 rows) revealed no single dominant source, but did reveal a strong joint concentration: `eg.ru` and `lenta.ru` together account for 77.3% of all clickbait examples, and 70.0% of all clickbait headlines fall under the `Общество` category, with fewer than 1% falling under `Происшествия` and only 2.7% under `Экономика`. Critically, the per-source specialisation was found to be near-total for two categories: every one of the 219 clickbait examples classified under `Общество` sourced from `lenta.ru` originates exclusively from that outlet's `Общество` coverage, and all 57 `Шоу-бизнес` clickbait examples originate exclusively from `eg.ru`. When the equivalent cross-tabulation was computed for the non-clickbait subset (696 rows), an inverse imbalance emerged for the `Происшествия` category: 98 of the 696 non-clickbait examples (14.1%) fall under `Происшествия`, concentrated overwhelmingly (84 of 98) in `aif.ru`, against only 5 clickbait examples (0.8%) in the same category — a seventeen-fold disparity that would allow a classifier to learn that crime-related lexical markers (“ДТП”, “пожар”, “задержан”) are near-perfect predictors of the non-clickbait class, independently of any genuine linguistic clickbait signal. Conversely, the `Шоу-бизнес` category exhibited the opposite pattern (57 clickbait rows versus only 4 non-clickbait rows), creating a comparable, inverse leakage risk around celebrity-related lexical markers.

Tables 3.4–3.6: Source–category leakage diagnostic (referenced in Section 3.2.2)

Source	Общество	Политика	Происшествия	Шоу-бизнес	Экономика	Row total
aif.ru	55	57	4	0	14	130
eg.ru	144	44	0	57	0	245
lenta.ru	219	0	0	0	0	219
mail.ru	2	1	1	0	2	6
Column total	420	102	5	57	16	600

Table 3.4: Source–category cross-tabulation, clickbait subset (N=600).

Source	Общество	Политика	Происшествия	Шоу-бизнес	Экономика	Row total
aif.ru	112	109	84	0	37	342
eg.ru	114	20	0	4	0	138
lenta.ru	150	0	0	0	0	150
mail.ru	28	17	14	0	7	66
Column total	404	146	98	4	44	696

Table 3.5: Source–category cross-tabulation, non-clickbait subset (N=696).

Category	Clickbait share	Non-clickbait share	Delta (pp)
Общество	420 (70.0%)	404 (58.0%)	+12.0 pp
Политика и Безопасность	102 (17.0%)	146 (21.0%)	−4.0 pp
Происшествия	5 (0.8%)	98 (14.1%)	−13.2 pp
Шоу-бизнес	57 (9.5%)	4 (0.6%)	+8.9 pp
Экономика	16 (2.7%)	44 (6.3%)	−3.7 pp

Table 3.6: Category distribution delta between clickbait and non-clickbait subsets. Flagged deltas (Происшествия, Шоу-бизнес) identify the high-leakage-risk categories pruned from the balanced non-clickbait variant (Section 3.4.2).

To mitigate this diagnosed risk without discarding any annotated clickbait examples — which, given the annotation effort invested, would have been costly — the two identified high-leakage categories were removed specifically from the *non-clickbait* subset when constructing the balanced corpus variant, reducing it from 696 to **594** rows (a reduction of 102 rows, corresponding exactly to the 98 Происшествия and 4 Шоу-бизнес rows identified as anomalous). Because the clickbait subset already exhibited a comparatively low a priori representation in these same two categories (5 and 57 rows respectively, together representing only 10.3% of the clickbait subset), it required no further category-level pruning and was

retained at its full annotated size of 600 rows for the corresponding balanced variant. Following this adjustment, the two dominant shared categories, Общество and Политика и Безопасность, jointly account for 87% of the clickbait subset and 79% of the (pruned) non-clickbait subset — a substantially more comparable topical footprint, intended to compel any classifier trained on this corpus to rely on the sentence-internal rhetorical and structural markers formalised in Section 2.2, rather than on topic- or source-level shortcuts. Source-level balance was independently verified as satisfactory without further pruning, as no source is exclusively associated with either class (e.g. aif.ru: 21.7% of the clickbait subset versus 49.1% of the non-clickbait subset — a difference, but not a degenerate one-sided association).

This leakage-diagnosis-and-mitigation procedure is documented here in detail, rather than relegated to an implementation footnote, because it constitutes direct evidence of the annotation-adjacent quality-control effort — beyond the annotation act itself — that the 90% figure cited at the opening of this chapter is intended to convey: constructing a resolution corpus is not merely a matter of writing summaries, but of actively auditing the resulting corpus for statistical artefacts that would undermine its validity as a training and evaluation resource.

3.2.3 Text cleaning

Prior to annotation, the raw headline and text fields extracted by the Stage 1 crawlers are passed through a dedicated cleaning routine that removes residual HTML tags (via the regular expression `<[>]+>`), decodes common HTML entities (e.g. ` `), strips control characters and non-breaking whitespace (` `, ` `, and other Unicode control ranges), and collapses repeated whitespace into single spaces. This step, implemented as `clean_text()` in `3_Анализ_датасета_TFM.ipynb` and applied to produce the file `Data_cut_parsed.csv`, ensures that the text subsequently presented to human annotators — and, later, to language models — is free of web-scraping artefacts that could otherwise interfere with both manual readability and automatic tokenisation.

3.2.4 Structural schema

The consolidated and cleaned corpus, `Data_cut_parsed.csv`, comprises **1,296 articles** (696 non-clickbait and 600 clickbait) and eleven columns: `url`, `source`, `language`, `category`, `headline`, `text`, `date`, `is_clickbait`, `clickbait_type`, `severity`, and `answer_summary`. This schema was deliberately designed to exceed the minimal requirements of the resolution task inherited from NoticIA (`headline`, `text`, `answer_summary`), for a specific reason of scope stated explicitly as a design goal from the outset of the project: to produce a corpus that is *structurally interoperable* with clickbait corpora constructed for other languages, and therefore reusable beyond the specific experiments reported in this TFM. Concretely:

- `url`, `source`, and `date` allow any future user of the corpus to trace provenance, verify claims against the original publication, and analyse temporal or outlet-level patterns — fields entirely absent from resolution-only schemas such as NoticIA's.
- `language` (fixed to Russian in the present release, but present as an explicit field rather than assumed implicitly) is included specifically to allow this corpus to be

concatenated with equivalently-structured corpora for other languages — including, in principle, NoticIA itself — into a multilingual clickbait-resolution resource without a schema migration step.

- category supports topical stratification and, as demonstrated in Section 3.2.2, is essential for auditing and mitigating source/category leakage — an analysis that would not have been possible under a minimal headline–text–answer_summary schema.

is_clickbait is a binary field populated for all 1,296 rows, while clickbait_type, severity, and answer_summary are populated conditionally: answer_summary is populated for **all** 1,296 rows (every article, clickbait or not, receives a concise summary — see Section 3.3), whereas clickbait_type and severity are populated exclusively for the 600 clickbait rows, in line with the ten-feature taxonomy defined in Section 2.2, which is by construction inapplicable to non-clickbait headlines. This conditional-population design was verified programmatically through a two-rule consistency check applied to the full 1,296-row corpus: Rule 1 confirms that no non-clickbait row (is_clickbait = 0) contains a non-null value in clickbait_type or severity; Rule 2 confirms that every clickbait row (is_clickbait = 1) contains a non-null value in both clickbait_type and answer_summary. Both rules were satisfied with zero violations across the full corpus, providing an auditable, reproducible guarantee of annotation-schema consistency that would not have been verifiable without the explicit schema design adopted here.

Column	Type	Population rule	Description
url	String	All rows	Article URL; supports provenance and traceability
source	String	All rows	News outlet (aif.ru, eg.ru, lenta.ru, mail.ru)
language	String	All rows	Fixed to “ru”; explicit field for multilingual interoperability
category	String	All rows	Harmonised topical category (Section 3.2.1)
headline	String	All rows	Raw article headline
text	String	All rows	Cleaned article body (Section 3.2.3)
date	Datetime	All rows	Publication date, YYYY-MM-DD HH:MM
is_clickbait	Int (0/1)	All rows	Binary clickbait label
clickbait_type	String	Clickbait rows only	Comma-separated taxonomy tags (Section 2.2.1)
severity	Float	Clickbait rows only	Raw weighted severity score (Section 2.2.2)
answer_summary	String	All rows ¹	Human-written one-sentence resolving summary

Table 3.7: Structural schema of *Data_cut_parsed.csv* ($N=1,296$).

¹ *answer_summary* is populated for 1,293 of 1,296 rows (99.8%); 3 non-clickbait rows from *aif.ru* lack a summary due to an annotation gap, discussed as a minor corpus-completeness caveat in Section 3.3.3.

3.3 Stage 3: Manual Annotation

3.3.1 Annotation protocol and scope

As stated in Section 1.3, the entirety of the corpus annotation reported in this work was carried out by a single annotator: the author of this TFM. Just as Noticia’s annotators were PhD students with native-level competence in Spanish (Section 2.1.2), the annotator in this work is a native speaker of Russian, ensuring the same level of linguistic and cultural familiarity with the target language’s clickbait conventions that underpinned the original Noticia annotation protocol. For each of the 1,296 articles in the consolidated corpus, the annotation task proceeds through the following steps, applied uniformly regardless of whether the article is ultimately judged to be clickbait:

- **Headline reading and question identification.** The annotator reads the headline in isolation, without yet consulting the article body, and asks: does this headline withhold information that a reader would need in order to fully understand its claim, and if so, what is the implicit question it poses? This step follows directly the annotation philosophy of Noticia (Section 2.1.2), in which annotators are instructed to first identify the question the headline poses to the reader before searching for its answer.
- **Clickbait judgement and feature scoring.** The annotator determines whether the headline exhibits clickbait characteristics and, if so, scores each of the ten binary features $p_1, \dots, p_{10}$ defined in Section 2.2.1 against the headline (and, where relevant to distinguishing a genuine content mismatch under the *BrokenPromise* feature, against the article body). The subset of features scored as present ($p_i=1$) determines the categorical tags recorded in the *clickbait_type* column (e.g. teaser, hype, vague), from which the *severity_score* and *severity_class* fields are subsequently derived automatically via the formula and calibrated thresholds established in Section 2.2.2–2.2.3.
- **Answer localisation and summary writing.** The annotator reads the full article body and locates the passage — frequently, though not universally, located towards the end of the article, consistent with the “buried lede” structure characteristic of clickbait journalism discussed in Section 1.1 — that resolves the question identified in Step 1. A single, concise, factual sentence is then written into the *answer_summary* field, closely mirroring the abstractive, single-sentence summarisation style established by the Noticia annotation guidelines (Section 2.1.2), while being independently authored against the Russian-language source material.

A methodologically significant difference from the Noticia protocol, already flagged in Section 3.2.4, is that this annotation step — Step 3 in particular — was applied not only to the

600 headlines ultimately judged as clickbait, but to **all 1,296 articles** in the corpus, including the 696 judged as ordinary, non-clickbait reporting. Every article in the corpus therefore carries a human-written `answer_summary`, regardless of its clickbait status, giving the resulting corpus a dual applicability that Noticia’s clickbait-only annotation does not offer: it can support not only clickbait-conditional resolution (train and evaluate a model to resolve a clickbait headline, as in Noticia) but also general-purpose one-sentence news summarisation, independently of the clickbait dimension, over the same 1,296-article corpus.

3.3.2 Single-annotator design: implications and mitigations

As acknowledged in Section 1.3, single-annotator annotation departs from Noticia’s double-annotation protocol, under which a second annotator independently produced validation summaries for a subset of the corpus, allowing García-Ferrero and Altuna (2024) to report a Spearman correlation of 0.99 between model rankings obtained against the two annotator sets — strong evidence of high inter-annotator reliability for the resolution task, but evidence that, by construction, cannot be independently replicated within a single-annotator design.

Three considerations partially mitigate, without eliminating, the resulting reliability risk in this work. First, the ten-feature taxonomy and weighted scoring formula developed in Section 2.2 were specifically designed to reduce the subjectivity of the clickbait-severity judgement: rather than requiring a single holistic “is this clickbait, and how bad” judgement, the annotator instead answers ten independently-specified, concretely-illustrated binary questions (Section 2.2.1), each accompanied by worked positive and negative examples during guideline development, which constrains the annotator’s discretion relative to an unstructured judgement. Second, the empirical recalibration of severity thresholds described in Section 2.2.3 — grounded in the observed distribution of tag co-occurrence rather than in an a priori assumption — provides an internal, data-driven consistency check on the annotator’s tagging behaviour: had the annotator applied the taxonomy inconsistently across the corpus, the resulting tag-count distribution would plausibly have appeared far noisier or less structured than the clear, monotonically decreasing 2-to-5-tag distribution actually observed (Table 2.2). Third, the schema-level consistency validation described in Section 3.2.4 (Rules 1 and 2) provides a further, fully auditable guarantee that the mechanical aspects of the annotation — field population — were applied without exception across all 1,296 rows.

These mitigations notwithstanding, this work explicitly does not claim inter-annotator reliability equivalent to Noticia’s validated 0.99 Spearman correlation, and this limitation, together with a concrete proposal for its future resolution through multi-annotator validation, is discussed critically in Section 6.2.

Figure 3.1

«Чёрная полоса Беллы Хадид»

eg.ru · Шоу-бизнес · tags: Hype, Teaser, Celeb

Вначале пошли разговоры о разрыве Беллы с бойфрендом — наездником Аданом Баньюэлосом. [...] Следом заговорили, что Адан арестован. Это оказалось правдой. [...]

А потом произошло уже настоящее несчастье: Хадид поделилась горем, что у нее умерла обожаемая такса Декстерн Ле Хойн. [...]

Severity score = 0.286 severity class = 2 (Medium)

answer_summary: «Сначала ложные слухи о расставание с парнем, затем его арест и смерть таксы Беллы»

Figure 3.1: Example of a fully annotated clickbait entry from the corpus, showing the categorical tags, continuous severity_score, discrete severity_class, and human-written answer_summary jointly derived for each headline (Section 2.2.4).

3.3.3 Descriptive statistics of the annotated corpus

A quantitative profile of the annotated corpus, computed in 3_Анализ_датасета_TFM.ipynb, is summarised in Table 3.8. Across the full 1,296-row corpus, headlines average 9.9 words, with a 95th-percentile length of 15 words — comfortably within the input-length budget of standard encoder architectures such as BERT-family models, were the corpus to be used for detection/classification purposes. Article bodies average 242.2 words, with a right-skewed distribution (median 173 words) reflecting a long tail of substantially longer investigative or listicle-style pieces, up to a maximum of 3,085 words — the single longest article in the corpus is, notably, a non-clickbait piece, consistent with the “buried lede” discussion below being about typical rather than extreme-case article length. The 95th-percentile length of 698 words already approaches or exceeds the practical context-length budget used for some of the models evaluated in Chapter 4, a consideration directly relevant to the model-selection rationale discussed in Section 4.1. answer_summary fields average 8.9 words (excluding three rows with no recorded summary, Table 3.7), with a tightly concentrated, near-symmetric distribution (median 9.0 words) and a maximum of 38 words, confirming that the single-sentence summarisation style targeted in Step 3 of the annotation protocol (Section 3.3.1) was applied consistently across the corpus.

Restricting this analysis to the clickbait subset alone reveals a systematic, and linguistically expected, divergence from the corpus-wide averages (Table 3.8): clickbait headlines are marginally longer than the corpus average (10.3 versus 9.9 words), consistent with the tendency of clickbait headlines to incorporate additional intriguing detail, emotionally charged modifiers, or embedded quotations (Section 2.2.1); and, more strikingly, the article bodies underlying clickbait headlines are themselves substantially longer than those underlying non-clickbait headlines (median 217 versus 156 words; mean 277.6 versus 211.6 words), with a markedly higher minimum length (52 words, versus single-sentence “stub” articles as short as 19 words appearing in the non-clickbait subset). This pattern is consistent with the “buried lede” structural hypothesis discussed in Section 1.1: clickbait articles tend to pad their content with filler material precisely in order to delay the delivery of the information promised by the headline, whereas ordinary, non-clickbait reporting more frequently delivers its core information concisely and without artificial elaboration.

Field	Subset	N	Mean	Median	Min	Max	P95
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Field	Subset	N	Mean	Median	Min	Max	P95
headline	Full corpus	1,296	9.9	10.0	2	20	15
	Clickbait	600	10.3	10.0	2	19	16
	Non-clickbait	696	9.5	9.0	3	20	14
text	Full corpus	1,296	242.2	173.0	19	3,085	698
	Clickbait	600	277.6	217.0	52	2,658	808
	Non-clickbait	696	211.6	156.0	19	3,085	552
answer_summary	Full corpus ¹	1,293	8.9	9.0	1	38	15
	Clickbait	600	9.1	9.0	1	38	16
	Non-clickbait ¹	693	8.7	9.0	1	30	14

Table 3.8: Length statistics (in words) of the annotated corpus, by field and by clickbait status.

¹ Excludes 3 non-clickbait rows (all from aif.ru) with no recorded answer_summary (Table 3.7).

Figure 3.2: Empirical distribution of tag co-occurrence per clickbait headline (N=600), reproducing Table 2.2.

3.4 Stage 4: Pre-processing for Model Consumption

3.4.1 Class-conditional file splitting

For downstream modelling purposes, the consolidated, cleaned, and annotated corpus Data_cut_parsed.csv (1,296 rows) is split along the is_clickbait field into two class-conditional files: Data_clickbait_1_full.csv (600 clickbait rows, retaining all eleven schema columns) and Data_clickbait_0_full.csv (696 non-clickbait rows). This split is the basis both for the leakage-diagnosis analysis described in Section 3.2.2 and for the balancing procedure described below.

3.4.2 Balanced variants and their motivation

As detailed in Section 3.2.2, a leakage-mitigated **balanced** variant of the non-clickbait file is produced by removing the two categories (Происшествия and Шоу-бизнес) identified as carrying disproportionate source/category signal, yielding Data_clickbait_0_balanced.csv (594 rows); the corresponding clickbait file, Data_clickbait_1_balanced_severity.csv, retains its full annotated size of 600 rows, augmented with the severity_score and severity_class fields computed as described in Section 2.2.3 (hence the _severity suffix), so that the balanced clickbait file can support severity-graded classification tasks directly, without requiring a separate join against the tagging data.

The purpose of producing this class-and-category-balanced pairing, beyond the leakage-mitigation rationale of Section 3.2.2, is to directly support a research use case explicitly targeted by the corpus design but outside the resolution-generation scope of NoticIA: training

and evaluating a binary or graded clickbait *classifier* under conditions that approximate a continuous, several-month stream of undifferentiated news headlines, as opposed to a comparison between a curated clickbait sample and an unrelated, thematically dissimilar control set. This directly operationalises the corpus-collection rationale set out in Section 3.1.1: because the underlying raw corpus was collected by comprehensively crawling ordinary news output rather than by sourcing pre-identified clickbait examples, the resulting balanced pairing constitutes, to the best of the author’s knowledge, a comparatively realistic proxy for the headline-scanning discrimination task faced by an actual reader — a property that a corpus built primarily from a clickbait-curation social media account, however efficient to construct, cannot by design offer.

3.4.3 Reduction to model-ready columns

Finally, for direct consumption by the language models evaluated in Chapter 4 — which require only the headline, article body, and target answer, and neither benefit from nor require the provenance and classification metadata retained for corpus-reuse purposes (Section 3.2.4) — each of the four class-conditional files (*_full* and *_balanced*, for both classes) is *reduced to a minimal three-column schema (headline, text, answer_summary)*, producing the four ready files used as the direct input to the few-shot and fine-tuning experiments described in Chapter 4: *Data_clickbait_1_balanced_ready.csv* (538 rows), *Data_clickbait_0_balanced_ready.csv* (594 rows), *Data_clickbait_1_full_ready.csv* (600 rows), and *Data_clickbait_0_full_ready.csv* (696 rows). The discrepancy between the 600-row *_full_ready* and 538-row *_balanced_ready* clickbait files reflects a further, model-facing filtering step applied during the few-shot evaluation pipeline itself, described in Section 4.2.

This four-stage pipeline — collection, structuring, annotation, and model-oriented pre-processing — together with the leakage-diagnosis procedure of Section 3.2.2 and the dual-purpose (resolution and detection) schema design of Section 3.2.4, constitutes the corpus construction methodology whose experimental exploitation is reported in Chapters 4 and 5.

4. Experimentación y Modelado

4.1 Experimental Design Rationale and Resource Constraints

Before describing the two experimental conditions evaluated in this chapter — few-shot in-context learning (Section 4.2) and cross-lingual fine-tuning (Section 4.3) — it is necessary to state explicitly the resource constraints under which this experimental phase was conducted, since they directly shaped several of the design decisions reported below, and since transparency about these constraints is itself part of the academic honesty this Memoria aims to uphold.

Unlike the original NoticIA evaluation, which was conducted on dedicated research infrastructure (four NVIDIA A100 80GB GPUs, Section 2.1.3) and whose zero-shot comparison spans over thirty models evaluated exhaustively across the full 850-article

dataset, the present work was conducted under the time and budget constraints characteristic of an individually executed TFM. Two concrete consequences follow from this:

First, **local inference of large open-weight models was not a viable option for models beyond a few billion parameters**. The consumer-grade GPU resources available through free-tier cloud notebooks (Google Colab T4 instances) impose a hard ceiling on the size of a model that can be loaded and run locally, even under aggressive 4-bit quantisation. Evaluating a 70-billion-parameter model such as Llama-3.1-70B-Instruct, or even the mid-sized Qwen3-VL-32B-Instruct, therefore required routing inference through a commercial third-party API aggregator (OpenRouter) rather than running the models locally.

Second, and consequently, **API-based inference on paid, metered models carries a direct monetary cost per token processed**, which — combined with the comparatively long input length of the articles in the corpus (Section 3.3.3) — made it economically and practically infeasible to run each model multiple times per test example (e.g. to average over several stochastic samples, or to conduct a systematic hyperparameter sweep over prompt variants for each of the four models) within the time and budget available for this TFM. Each of the four few-shot models evaluated in Section 4.2 was therefore run **once**, at a low, near- deterministic decoding temperature, over the full evaluation split, rather than multiple times with results averaged — a methodological simplification relative to best practice in LLM evaluation that is acknowledged here explicitly, and revisited as a concrete direction for future work in Section 6.2. This constraint also directly informed the selection of the four evaluated models (Section 4.2.1): the pool was deliberately restricted to a small, fixed set of models runnable within budget, rather than the broad, several-dozen-model sweep undertaken by Noticia on dedicated institutional compute.

These constraints do not, in the author’s assessment, invalidate the comparative conclusions drawn in Chapter 5 — the four few-shot models are evaluated under identical conditions relative to one another (Section 4.2.2), which is the property that matters for a *relative* comparison — but they do mean that the reported metrics should be read as a single-pass snapshot of model behaviour rather than as a statistically averaged estimate of expected performance, and that the experimental scope of this chapter is narrower than that of Noticia’s original evaluation. This is stated here plainly, in keeping with the principle that a smaller, honestly-scoped experiment is preferable to an overstated one.

4.2 Few-shot In-Context Learning Evaluation

4.2.1 Model selection

Four instruction-tuned language models, spanning a range of parameter scales and model families, were selected for few-shot evaluation, all accessed through a single, unified inference interface (the OpenRouter API, using the OpenAI-compatible chat-completions endpoint):

Model identifier	Approx. scale	Provenance
------------------	------------------	------------

Model identifier	Approx. scale	Provenance
qwen/qwen-2.5-72b-instruct	72B	Alibaba / Qwen (Qwen Team, 2024)
qwen/qwen3-vl-32b-instruct	32B	Alibaba / Qwen, Qwen3 generation (Qwen Team, 2025)
qwen/qwen-2.5-7b-instruct	7B	Alibaba / Qwen (Qwen Team, 2024)
meta-llama/llama-3.1-70b-instruct	70B	Meta (Grattafiori et al., 2024)

This selection intentionally spans both a within-family scale comparison (Qwen2.5-7B versus Qwen2.5-72B, holding model family and training recipe approximately constant while varying scale by an order of magnitude) and a cross-family comparison at comparable scale (Qwen2.5-72B versus Llama-3.1-70B), together with one model drawn from the newer Qwen3 generation (Qwen3-VL-32B) included to probe whether generational improvements in instruction-tuning and multilingual pre-training data provide an advantage independent of raw parameter count. As discussed in Section 4.1, this pool is smaller than the exhaustive sweep conducted by Noticia; it was sized specifically to remain tractable under the time and API-cost constraints of this project while still supporting a scale-and-family comparison analogous in spirit, if not in breadth, to the one reported in Section 2.1.2.

4.2.2 Unified prompt design: equal conditions across models

The central methodological principle governing this evaluation, and the aspect of the experimental design given the greatest attention in this work, is that **all four models were evaluated under an identical prompt template, an identical single worked demonstration example, and identical decoding conditions**, in order to isolate model capability as the only varying factor between runs and to support a fair, controlled comparison. Concretely, for every test instance and for every one of the four models, the model receives an identical three-turn conversation:

- A **system instruction**, fixed verbatim across all four models, specifying the task in Russian: the model is instructed to act as a clickbait-resolution expert, to read the headline and article body, and to output a single, short, factually grounded sentence (or short phrase) that resolves the intrigue posed by the headline, strictly in Russian, without introductory phrases (e.g. “Ответ:”) or greetings.
- A single **worked demonstration** (one-shot, rather than the multi-shot demonstrations more commonly reported in the few-shot learning literature — see the discussion of this trade-off in Section 2.3.3): a fixed, domain-general example headline about programming languages, paired with a short article excerpt and its target answer (“Python и JavaScript”), presented as a user–assistant exchange preceding the actual test instance.
- The **test instance** itself: the target headline and article body, formatted identically to the demonstration turn.

This single, fixed demonstration example was deliberately chosen from a neutral, non-political, non-emotionally-charged domain (technology/career advice) and deliberately kept constant across every model and every test instance, rather than being tailored per model or per example, precisely in order to prevent any model from receiving an inadvertent advantage through a more favourable or more topically related demonstration. All four models were queried with a low decoding temperature (0.1), chosen to favour deterministic, low-variance output appropriate for a single-pass evaluation under the resource constraints discussed in Section 4.1, and API-level failures (e.g. rate-limiting or transient service errors) were handled through a retry-with-backoff mechanism, with any input that exhausted its retry budget recorded with an explicit error marker rather than silently dropped, so that failure cases remain auditable in the final prediction file rather than being invisibly excluded from the evaluation.

Figure 4.1: Unified three-turn few-shot prompt template, applied identically across all four evaluated models (Section 4.2.2).

The evaluation was conducted on the Data_clickbait_1_balanced_ready.csv split described in Section 3.4.3, comprising clickbait headline–body–answer triplets only, consistent with the resolution-task framing of NoticIA (Section 2.1.2): the few-shot evaluation reported here concerns the model’s ability to *resolve* a headline already known to be clickbait, rather than to jointly detect and resolve it. A small number of rows were excluded from the final aggregated comparison due to persistent API-level failures recorded during collection (see Section 4.1), reducing the working evaluation set from the full 600-row clickbait annotation to **538 rows**, consolidated into the single comparison file FINAL_ALL_MODELS_PREDICTIONS_538.csv, which pairs the original headline, article text, and gold-standard answer_summary with the four models’ predictions (Predicted_Answer_Qwen72B, Predicted_Answer_Qwen3VL_32B, Predicted_Answer_Qwen7B, Predicted_Answer_Llama31_70B) in a single row per test instance. This consolidated file constitutes the direct input to the evaluation protocol and comparative analysis reported in Chapter 5.

Table 4.1

Example A — strong agreement

Headline	«Академик назвал среднюю продолжительность жизни в России»
Gold	Средняя продолжительность жизни в России составляет 74,2 года
Qwen2.5-72B	74,2 года
Qwen3-VL-32B	Средняя продолжительность жизни в России составляет 74,2 года.
Qwen2.5-7B	74,2 года
Llama-3.1-70B	74,2 года

Example B — strong divergence

Headline	«Жителей Москвы предупредили о ненормальной погоде в декабре»
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Headline	«Жителей Москвы предупредили о ненормальной погоде в декабре»
Gold	В Москве декабрь будет аномально теплым, а зима наступит позже
Qwen2.5-72B	Аномально теплый декабрь, снег и морозы только во второй половине месяца
Qwen3-VL-32B	В декабре 2025 года в Москве ожидается аномальная теплая погода с задержкой настоящей зимы до второй половины месяца.
Qwen2.5-7B	Теплая и снежная зима опоздает в Москве
Llama-3.1-70B	Минус 8,6 градусов.

Table 4.1: Representative examples from the consolidated few-shot prediction file (FINAL_ALL_MODELS_PREDICTIONS_538.csv), illustrating a case of strong cross-model agreement (A) and a case of strong divergence, including one clearly hallucinated prediction (B).

4.2.3 A note on the departure from NoticIA’s zero-shot protocol

It is worth noting explicitly that NoticIA’s own evaluation (Section 2.1.2) adopted a strictly zero-shot protocol rather than a few-shot one, a choice its authors justify by observing that the average article length in their dataset (approximately 550 words) makes few-shot prompting — which would require embedding one or more full demonstration articles — impractical within the context-length and cost budgets available to them. The present work departs from this precedent by adopting a one-shot protocol instead, made tractable specifically because the single demonstration example used (Section 4.2.2) employs a short, synthetic article excerpt rather than a full-length news article, keeping the added context-length and cost overhead of the demonstration turn comparatively small relative to the average 555.8-token article body reported in Section 3.3.3. This departure was adopted, rather than NoticIA’s zero-shot design, specifically because — as discussed in Section 2.3.3 — in-context demonstration was expected to provide the models with an explicit, unambiguous illustration of the required output format (single short sentence, no introductory phrase, strictly in Russian), a formatting requirement that, in preliminary informal testing, proved harder to enforce through instruction alone than through a combination of instruction and demonstration.

4.3 Fine-tuning ClickbaitFighter-7B: An Exploratory Cross-lingual Adaptation Experiment

4.3.1 Framing: an exploratory experiment, not a production training run

The second experimental condition evaluated in this work is the cross-lingual fine-tuning of ClickbaitFighter-7B (Section 2.1.3) on the Russian-language annotated corpus. It is important to frame this experiment accurately, in line with the resource constraints discussed in Section 4.1: unlike the original ClickbaitFighter training runs, which were conducted with full-parameter fine-tuning under bfloat16 precision on four dedicated NVIDIA A100 80GB GPUs (Section 2.1.3), the fine-tuning experiment reported here was conducted on a single, free-tier

NVIDIA T4 GPU (16GB), a hardware environment that categorically precludes full-parameter fine-tuning of a 7-billion-parameter model. This experiment is therefore best understood, and is presented in this Memoria, not as a rigorously optimised training pipeline intended to reproduce or exceed the original ClickbaitFighter-7B’s Spanish-language performance in a new language, but as an **exploratory feasibility experiment**: an attempt to establish, under realistic hobbyist/ single-researcher hardware constraints, whether targeted, parameter- efficient adaptation of an existing Spanish-language task-specific model is capable of producing measurable, qualitatively coherent Russian- language clickbait-resolution behaviour at all — a first empirical data point on the cross-lingual transfer question motivated in Section 2.3.4, rather than a finished, production-grade Russian ClickbaitFighter model.

4.3.2 Parameter-efficient adaptation via QLoRA

To make fine-tuning tractable on a single T4 GPU, the experiment adopts QLoRA-style (Dettmers et al., 2023) parameter-efficient fine-tuning rather than full-parameter fine-tuning: the base Iker/ClickbaitFighter-7B model is loaded in 4-bit precision (NF4 quantisation, double quantisation enabled, float16 compute dtype) and prepared for k-bit training, after which a Low-Rank Adaptation (LoRA) adapter is injected into all major linear projection layers of the transformer (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj), with rank r , scaling factor α , and a dropout of 0.05 applied to the adapter layers. Under this configuration, only the injected low-rank adapter matrices are updated during training, while the base model’s original weights remain frozen in their quantised form — a design that reduces the number of trainable parameters by roughly two orders of magnitude relative to full-parameter fine-tuning, at the cost of a strictly smaller hypothesis space for adaptation than the full-parameter fine-tuning protocol used for the original Spanish ClickbaitFighter models. $r=16, \alpha=32$

Training data is drawn from Data_clickbait_1_balanced_ready.csv (538 clickbait headline–text–answer triplets, Section 3.4.3), formatted through the model’s native chat template as a three-turn exchange (system instruction, user turn containing the headline and article body, assistant turn containing the gold answer_summary), and tokenised with truncation to a maximum sequence length of 1,024 tokens — substantially shorter than the 8,192-token sequence length used in the original ClickbaitFighter training (Section 2.1.3), a further consequence of the memory constraints of the available hardware, and one that, given the article-length distribution reported in Section 3.3.3 (95th percentile of 1,580 tokens), entails that the longest articles in the training set are partially truncated during fine-tuning. Training is conducted for 2 epochs with a per-device batch size of 2 and gradient accumulation over 4 steps (yielding an effective batch size of 8), a learning rate of 2×10^{-4} , fp16 mixed-precision arithmetic, and the memory- efficient paged_adamw_8bit optimiser — all choices directly reflecting the constrained-hardware framing of Section 4.3.1 rather than an independently optimised training recipe.

4.3.3 Iterative tuning of inference-time decoding parameters

While the LoRA training configuration itself (Section 4.3.2) was fixed prior to training, obtaining coherent, non-repetitive, appropriately concise output from the fine-tuned adapter at

inference time required substantial iterative experimentation. Over the course of this experiment, more than ten distinct configurations of the generation (decoding) parameters were trialled against a small held-out set of test predictions before arriving at a configuration judged qualitatively acceptable, in order to address two recurring failure modes observed during early trials: (i) degenerate repetition loops, in which the model would repeat a phrase or clause multiple times within a single output, and (ii) premature truncation, in which the generated answer was cut off mid-sentence before completing a coherent thought. The final decoding configuration adopted — sampling-based generation (`do_sample=True`) with temperature 0.4, nucleus sampling with `top_p=0.85`, a repetition penalty of 1.2, and an explicit `no_repeat_ngram_size` of 3 — reflects this iterative, empirically-driven tuning process rather than a value adopted from a single principled derivation, and is reported here as such: as an example of the kind of practical, trial-and-error engineering effort that a resource-constrained, single-researcher fine-tuning experiment requires, in contrast to the systematic hyperparameter protocols achievable with the dedicated compute budget available to the original ClickbaitFighter training effort (Section 2.1.3). A secondary, input-side mitigation was also required to keep single-example inference within the memory budget of the T4 GPU: article bodies exceeding approximately 1,200 characters are truncated prior to being passed to the fine-tuned model at inference time, a pragmatic safeguard against out-of-memory failures that, as with the training-time sequence-length truncation discussed in Section 4.3.2, disproportionately affects the longest articles in the corpus.

4.3.4 Output of the experiment

The fine-tuned LoRA adapter, saved independently of the frozen base model’s weights, is subsequently reloaded and merged with the quantised base model at inference time to generate predictions over the same 538-row evaluation split used for the few-shot comparison in Section 4.2.2, enabling the fine-tuned model’s output to be evaluated against the same gold-standard `answer_summary` references and under the same automatic-metric protocol (BLEU, ROUGE-L, average answer length) defined in Chapter 5, alongside — rather than as a categorically separate exercise from — the four few-shot models. The comparative results of this five-system evaluation (four few-shot models and one exploratory fine-tuned adaptation) are reported and discussed in Chapter 5.

5. Evaluación y Análisis de Resultados

5.1 Evaluation Metrics

Following the evaluation protocol established by NoticiaIA (Section 2.1.2) and standard practice in the automatic summarisation and text-generation literature, three complementary metrics are used to compare each system’s generated answer against the single human-written gold `answer_summary`: **BLEU**, **ROUGE-L**, and the **average length**, in words, of the generated answers. This section defines each metric formally and states the exact configuration used in this work, since, as discussed below, seemingly minor configuration

choices (e.g. which BLEU variant, which smoothing method) materially affect the resulting scores and must therefore be made explicit for the results to be interpretable and reproducible.

5.1.1 BLEU

BLEU (Bilingual Evaluation Understudy; Papineni et al., 2002) is a precision-oriented metric originally developed for machine translation evaluation, which measures the overlap of contiguous n-gram sequences between a candidate (generated) text and one or more reference texts. For a given n-gram order n , the modified n-gram precision p_n is computed as the number of n-grams in the candidate that also occur in the reference (each reference n-gram counted at most once, to prevent degenerate repetition from inflating the score), divided by the total number of n-grams in the candidate. The standard BLEU score aggregates precisions across multiple n-gram orders (typically $n=1,\dots,4$) via a geometric mean, modulated by a brevity penalty (BP) that penalises candidates shorter than the reference:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_n w_n \log p_n\right)$$

$$\text{BP} = 1, \text{ if } c > r$$

$$\text{BP} = \exp(1 - r/c), \text{ if } c \leq r$$

where c is the candidate length, r the reference length, and w_n the weight assigned to each n-gram order (typically uniform, $w_n=1/N$).

In this work, given the characteristically short, single-sentence nature of the target answers (average 9.1 words in the full corpus, Section 3.3.3), a **unigram-only BLEU-1 variant** is adopted ($N=1$, i.e. $w_1=1$, all higher-order weights set to zero), computed at the word level rather than the sub-word or character level, using NLTK’s `sentence_bleu` implementation. Because unigram overlap on such short sequences frequently yields a raw precision of exactly zero whenever the candidate happens to share no single word with the reference — an outcome that would otherwise produce an uninformatively harsh, all-or-nothing score for many rows — a smoothing function (NLTK’s `SmoothingFunction.method1`, which adds a small constant to zero-count n-gram matches) is applied throughout, ensuring that partial lexical overlap is still rewarded on a continuous scale rather than collapsing all zero-overlap candidates to an identical floor score. This unigram, smoothed configuration is a deliberate simplification relative to the standard 4-gram BLEU most commonly reported in the machine-translation literature, motivated by the short target length in this task; it is reported here as such, and the reader should bear in mind that scores under this configuration are not directly numerically comparable to 4-gram BLEU scores reported elsewhere in the literature, including Noticia, which reports ROUGE-L only rather than BLEU.

5.1.2 ROUGE-L

ROUGE-L (Lin, 2004) is a recall-oriented metric, widely adopted in the summarisation literature (and the primary metric used by Noticia, Section 2.1.2), based on the length of the **Longest Common Subsequence** (LCS) between the candidate and reference sequences. Unlike n-gram-based metrics such as BLEU, the LCS need not be contiguous, allowing ROUGE-L to reward candidates that preserve the reference’s word *order* without requiring an

unbroken matching span, which makes it comparatively more tolerant of minor insertions or reorderings than exact n-gram overlap. Given a candidate X of length m and a reference Y of length n, ROUGE-L is computed as the F-measure over LCS-based precision and recall:

$$R_{\text{LCS}} = \text{LCS}(X, Y) / n$$

$$P_{\text{LCS}} = \text{LCS}(X, Y) / m$$

$$F_{\text{LCS}} = (1 + \beta^2) \cdot R_{\text{LCS}} \cdot P_{\text{LCS}} / (R_{\text{LCS}} + \beta^2 \cdot P_{\text{LCS}})$$

with $\beta=1$ throughout this work, yielding the standard ROUGE-L F1 score reported below. For the exploratory fine-tuned model evaluated in Section 5.3, the complementary ROUGE-1 and ROUGE-2 F1 scores (based on unigram and bigram overlap respectively, using the same F-measure construction as above) are additionally reported, following the fuller diagnostic protocol applied to that system.

5.1.3 Average answer length

Finally, following the explicit rationale established by NoticIA (Section 2.1.2) — that, all else being equal, a *shorter* answer that still resolves the headline’s implicit question is preferable to a longer one, since excessive length undermines the “ultrasummary” objective of the resolution task (Section 1.1) — the average length, in words, of each system’s generated answers is reported and compared against the average length of the human-written gold answers (9.1 words across the full annotated corpus, Section 3.3.3; 8.79 words within the specific 538-row evaluation subset used in this chapter, Section 4.2.2). A system that achieves a comparatively strong BLEU/ROUGE-L score while also producing answers close in length to the human reference is, on these grounds, judged more favourably than one achieving a similar score through verbose, padded output — a consideration returned to directly in the analysis of Section 5.2.

5.2 Comparative Results: Few-shot Models

Table 5.1 reports the three metrics defined in Section 5.1, computed over the full 538-row evaluation set (FINAL_ALL_MODELS_PREDICTIONS_538.csv, Section 4.2.2), for each of the four few-shot models evaluated under the unified prompt protocol of Section 4.2.2.

Model	BLEU-1	ROUGE-L (F1)	Avg. length (words)
Qwen2.5-72B-Instruct	0.1704	0.2043	8.27
Qwen3-VL-32B-Instruct	0.1673	0.1924	18.31
Qwen2.5-7B-Instruct	0.1470	0.1769	7.26
Llama-3.1-70B-Instruct	0.1227	0.1605	7.11
<i>Human gold reference</i>	—	—	8.79

Table 5.1: Comparative few-shot evaluation results, 538-row evaluation set.

Figure 5.1: ROUGE-L score versus average answer length for the four few-shot models, with the human gold-reference length marked.

5.2.1 Ranking and the role of model scale

The four models exhibit a consistent ranking across both BLEU-1 and ROUGE-L: **Qwen2.5-72B-Instruct** > **Qwen3-VL-32B-Instruct** > **Qwen2.5-7B-Instruct** > **Llama-3.1-70B-Instruct**. This ranking offers two points of direct continuity with the discussion in Section 2.1.2, where it was noted that Noticia’s own zero-shot results showed model scale to be a comparatively weak predictor of resolution quality relative to the composition and quality of a model’s instruction-tuning data. The present results echo this pattern in one respect and complicate it in another. On one hand, the two Qwen-family models occupy the top two positions despite differing in scale by more than a factor of two (72B versus 32B), suggesting that family-level training recipe and Russian-language pre-training coverage may matter more, for this specific task, than raw parameter count within the family. On the other hand, and more strikingly, Llama-3.1-70B-Instruct — the second-largest model evaluated — obtains the lowest score of the four**, underperforming even the substantially smaller Qwen2.5-7B-Instruct (a tenfold difference in parameter count). Taken together with the Noticia finding that Llama-2-70b-chat-hf similarly underperformed the smaller OpenChat-3.5-0106 on the equivalent Spanish-language task (Section 2.1.2), this result lends further, now cross-lingual, support to the broader claim that general-purpose model scale alone does not reliably predict clickbait-resolution quality, and that family-specific factors — plausibly, differences in the proportion and quality of Russian-language and instruction-following data included during each family’s pre-training and alignment process — dominate the comparison instead. This work does not have access to the proprietary training data compositions of the evaluated models and can therefore only report, rather than causally explain, this association; it is noted here as an empirical observation consistent with, and extending to Russian, a pattern already documented for Spanish.

5.2.2 The role of answer length

A second, methodologically instructive pattern emerges when the length column of Table 5.1 is read alongside the score columns. **Qwen2.5-72B-Instruct** produces answers whose average length (8.27 words) is the closest of the four models to the human gold-reference length (8.79 words) within this evaluation subset, and simultaneously achieves the highest BLEU-1 and ROUGE-L scores — consistent with the general expectation that, for a fixed underlying semantic accuracy, an answer whose length distribution matches the reference will tend to score more favourably under overlap-based metrics, both because a well-calibrated length reduces the risk of the ROUGE-L precision term being diluted by superfluous generated words, and because it avoids the BLEU brevity penalty being triggered from the opposite direction. **Qwen3-VL-32B-Instruct**, by contrast, produces markedly longer answers (18.31 words, more than double the gold-reference length) yet still achieves the second-highest score of the four — indicating that this model’s comparatively verbose output nonetheless retains a high degree of genuine lexical and sequential overlap with the reference, rather than score inflation through length alone (were length alone driving the score, ROUGE-L’s recall-oriented construction, Section 5.1.2, would in principle be expected to reward — not penalise — additional generated words that happen to overlap with the reference, but ROUGE-L’s F-measure construction jointly penalises the corresponding drop in precision

from excess non-matching content, so the fact that Qwen3-VL-32B’s score remains high despite its verbosity is itself evidence of comparatively strong content selection). Conversely, both **Qwen2.5-7B-Instruct** and **Llama-3.1-70B-Instruct** produce answers shorter, on average, than the gold reference (7.26 and 7.11 words respectively) while achieving the two lowest scores — indicating that, for these two models, brevity is not compensating for a comparative shortfall in identifying and reproducing the specific lexical content of the correct answer. Taken together, these observations support the conclusion that answer length alone is an insufficient explanation for the observed ranking, and that genuine differences in the models’ Russian-language reading-comprehension and answer-localisation capability (Section 2.3.2) are the more plausible primary driver of the score differences reported in Table 5.1.

5.3 Results: Cross-lingual Fine-tuned Model (ClickbaitFighter-7B-RU)

Table 5.2 reports the results of the exploratory fine-tuning experiment described in Section 4.3, evaluated over the same 538-row evaluation set used for the few-shot comparison in Section 5.2, using the fuller diagnostic metric set (ROUGE-1, ROUGE-2, and ROUGE-L, alongside BLEU-1 and average length) applied specifically to this system.

Metric	Value
N (evaluation rows)	538
ROUGE-1 (F1)	0.0199
ROUGE-2 (F1)	0.0073
ROUGE-L (F1)	0.0193
BLEU-1	0.0088
Avg. length, gold reference (words)	9.5
Avg. length, model output (words)	25.7

Table 5.2: Evaluation results for the cross-lingual fine-tuned ClickbaitFighter-7B-RU model.

5.3.1 Interpreting a markedly lower score

The scores reported in Table 5.2 are, on every metric, substantially lower than any of the four few-shot models reported in Table 5.1 — by roughly an order of magnitude on ROUGE-L and BLEU-1. Read naively, this might suggest that the fine-tuning experiment failed outright to produce a usable clickbait-resolution model. This work argues, however, that such a reading would be both an oversimplification and, in one specific respect, a misapplication of the evaluation metrics themselves, for four compounding reasons that must be considered jointly rather than in isolation.

First, and most directly visible in Table 5.2 itself, is a pronounced length mismatch. The fine-tuned model’s output averages 25.7 words per answer — approximately **2.7 times longer** than the 9.5-word gold reference for this system’s evaluation subset, and roughly three times longer than any of the four few-shot models’ outputs (Table 5.1). Both BLEU and ROUGE-L are, by construction (Section 5.1.1–5.1.2), highly sensitive to length mismatch: BLEU’s precision term is computed over the candidate’s own n-grams, so a threefold increase in

candidate length mechanically dilutes the proportion of those n-grams that can possibly match a short, 9-word reference, even under otherwise perfect content selection; ROUGE-L’s F-measure construction similarly penalises excess candidate length through its precision component. This length inflation is a direct, traceable consequence of the resource-constrained fine-tuning setup documented in Section 4.3: the model was trained under a truncated 1,024-token sequence length (against the original ClickbaitFighter’s 8,192, Section 4.3.2) and required more than ten rounds of inference-time decoding-parameter adjustment specifically to mitigate — without fully eliminating — a tendency toward verbose, repetition-adjacent output (Section 4.3.3); the residual verbosity visible in the 25.7-word average is consistent with that mitigation having been only partially successful.

Second, and this is the point requiring the greatest interpretive care, BLEU and ROUGE-L are lexical-overlap metrics: they are fundamentally insensitive to correct answers expressed through synonymy, paraphrase, or morphological variation, a limitation that is disproportionately consequential for a morphologically rich, free-word-order language such as Russian (Section 1.2). Both metrics credit a candidate answer only to the extent that it reproduces the *exact surface word forms* of the single available human reference; a model that correctly identifies and communicates the same underlying fact using a different, equally valid lexical or grammatical realisation — for instance, a different but semantically equivalent verb, a synonymous noun, or an alternative case-inflected form of the same lemma, all common and idiomatic in Russian — receives no credit for that content under either metric, despite having correctly resolved the headline. Because this work’s corpus, unlike NoticIA’s (which, as discussed in Section 2.1.2 and Section 3.3.2, benefits from a second annotator’s independent reference for at least a validation subset, allowing evaluation against multiple acceptable phrasings), provides only a **single** human-written reference per headline (Section 3.3.2), this synonymy-blindness is not mitigated by averaging against multiple references, as is common practice in machine-translation evaluation, and is expected to compound with the length-mismatch effect described above to further suppress the reported scores below what a qualitative reading of the model’s output would suggest.

Third, informal inspection of the fine-tuned model’s raw output during the iterative decoding-parameter tuning described in Section 4.3.3 revealed a recurring artefact: the model would, on a subset of generations, intersperse Latin-alphabet characters or sub-word fragments within otherwise Cyrillic Russian words**** — a plausible consequence of the aggressive 4-bit quantisation and constrained adaptation capacity of the LoRA configuration (Section 4.3.2) interacting with the base model’s originally Spanish-and-English-dominant sub-word vocabulary under distributional shift toward Cyrillic script. Any token-level corruption of this kind directly and mechanically reduces exact n-gram and LCS matches against a cleanly-typed Cyrillic reference, independently of whether the underlying semantic content of the answer was otherwise correct, further compounding the two effects described above.

Fourth, and returning to the framing established in Section 4.3.1, these results must be read against the resource constraints under which this fine-tuning experiment was conducted: a single T4 GPU, 4-bit quantised QLoRA adaptation rather than full-parameter fine-tuning,

truncated training sequences, and a training set of 538 examples — a fraction of the compute budget (four A100 80GB GPUs, full-parameter fine-tuning, 8,192-token sequences) available to the original ClickbaitFighter training effort (Section 2.1.3). The low absolute scores reported in Table 5.2 are, in the author’s assessment, better interpreted as an expected outcome of this specific, resource-constrained exploratory configuration than as evidence that cross-lingual adaptation of ClickbaitFighter to Russian is not a viable direction in principle; this distinction, and the concrete changes to the fine-tuning protocol that would be required to test that broader claim properly, are discussed as future work in Section 6.2.

Table 5.3

(a) Verbosity + partial synonymy

Headline	<i>«Не самая малая Азия. Почему Малазия так охотно развивает отношения с РФ»</i>
Gold	Из-за ресурсов и защиты
CBF-7B-RU	Из-за богатых природных ресурсов и возможностей для бизнеса, особенно в строительстве и транспортном секторе. Также общая геополитическая ситуация в [...]

(b) Correct paraphrase, different wording

Headline	<i>«В США женщине диагностировали деменцию из-за надкусанных продуктов»</i>
Gold	Болезнь диагностировали на основании забывчивого поведения с едой
CBF-7B-RU	Деменция была диагностирована из-последствия неосторожной поведения (надкусания продукта) и неправильного обращения с кухонным оборудованием. [...]

(c) Latin-script character mixing artefact

Headline	<i>«Всплыла главная тайна Макрона: что будет с главой Франции после скандала»</i>
Gold	нет чёткой информации
CBF-7B-RU	Упоминается только одно слово “boyfriend”, которое может относиться ко всему спектру отношений, включая дружбу. Раскрыто ничего нового. [...]

Table 5.3: Qualitative examples from the fine-tuned model (predictions_7B_clickbait.csv), selected to illustrate cases where the output is semantically reasonable, or interpretable, despite a low automatic score, substantiating the discussion in Section 5.3.1.

5.4 Synthesis: Comparing All Five Systems

Bringing together Table 5.1 and Table 5.2, a joint ranking across all five evaluated systems (four few-shot models and one exploratory fine-tuned model), ordered by ROUGE-L F1, is:

Qwen2.5-72B-Instruct (0.2043) > Qwen3-VL-32B-Instruct (0.1924) > Qwen2.5-7B-Instruct (0.1769) > Llama-3.1-70B-Instruct (0.1605) >> ClickbaitFighter-7B-RU (0.0193).

This ordering stands in direct contrast to the corresponding result reported for Spanish in Noticia (Section 2.1.3), where the fine-tuned, task-specific ClickbaitFighter models markedly *outperformed* every evaluated general-purpose zero-shot model, including models an order of magnitude larger. The present work’s cross-lingual adaptation experiment does not reproduce that advantage for Russian — but, per the discussion in Section 5.3.1, this divergence is attributable, in the first instance, to the vastly different resource envelopes of the two fine-tuning efforts (Section 4.3.1, Section 2.1.3) and to specific, identifiable artefacts of this experiment’s constrained configuration (length inflation, Section 5.3.1; residual repetition tendencies, Section 4.3.3; occasional script-mixing, Section 5.3.1), rather than to a principled finding that cross-lingual task-specific adaptation is intrinsically less promising than general-purpose few-shot prompting for this task. The central, cautious empirical conclusion this work draws from the joint comparison is therefore not that fine-tuning underperforms few-shot prompting for Russian clickbait resolution as a matter of principle, but that **the specific exploratory fine-tuning configuration evaluated in this TFM, conducted under severe hardware constraints, did not close — and, under the metrics available, appears to widen — the gap that the equivalent, properly-resourced experiment closed for Spanish**; whether a better-resourced replication of this experiment (full-parameter fine-tuning, longer sequence lengths, a larger and multiply-annotated training set) would narrow or reverse this gap remains an open question, proposed as a direct avenue for future work in Section 6.2.

A secondary observation, consistent across both few-shot and fine-tuned conditions, is that the models evaluated in this chapter — irrespective of paradigm — did not reach the near-human ROUGE-L performance range documented for the best Spanish few-shot and fine-tuned systems in Noticia (Llama-3-70B-Instruct at 36.30, ClickbaitFighter-7B at 49.81 against a human baseline of 55.98, all measured on a different metric scale and evaluation subset, and therefore not directly numerically comparable to the results in this chapter, but informative as a broad point of reference). This overall gap is consistent with, and does not contradict, the central research-gap motivation set out in Section 1.2: Russian-language clickbait resolution remains a comparatively under-served capability across current general-purpose and task-specific models alike, reinforcing — rather than undermining — the value of the dedicated Russian-language corpus constructed in Chapter 3 as a resource for future, better-resourced modelling efforts on this task.

6. Conclusiones y Trabajos Futuros

6.1 Conclusions

This TFM set out, as stated in Section 1.4, to design, construct, and validate a Russian-language corpus for clickbait characterisation and resolution, and to empirically evaluate current-generation LLMs — under few-shot prompting and under targeted cross-lingual fine-tuning — on the resulting resource. Revisiting the nine specific objectives enumerated in

Section 1.4 in light of the work reported in Chapters 3 through 5, the following conclusions can be drawn.

On corpus construction (Objectives 1–5), which this work maintains throughout as its central methodological achievement. A four-source web crawling pipeline was designed and implemented (Section 3.1), yielding a raw corpus of 2,427 unique Russian-language news articles collected through comprehensive, undifferentiated sampling of ordinary news output — rather than through a pre-curated clickbait source, as in Noticia (Section 2.1.2) — a design choice made specifically to support a corpus representative of the continuous, undifferentiated headline stream an actual reader encounters (Section 3.1.1). From this raw yield, 1,296 articles were consolidated, cleaned, and structured into an eleven-column schema (Section 3.2.4) deliberately designed for cross-linguistic interoperability with resolution corpora built for other languages, including Noticia itself. Of these, 600 were annotated as clickbait, following the formalised, empirically-recalibrated ten-feature taxonomy and severity-scoring formula developed specifically for this work (Section 2.2) — a graded characterisation layer that Noticia’s binary annotation protocol does not offer — and a systematic source/category cross-tabulation analysis (Section 3.2.2) was conducted and acted upon to actively mitigate shortcut-learning risk in the resulting corpus, reducing the non-clickbait subset from 696 to a leakage-mitigated 594 rows. Every one of the 1,296 articles, clickbait or not, was additionally annotated with a human-written, single-sentence answer_summary (Section 3.3.1), giving the resulting resource a dual applicability — clickbait-conditional resolution and general-purpose one-sentence summarisation — that extends beyond the resolution-only scope of its primary methodological antecedent. As stated repeatedly throughout this Memoria, this corpus-construction effort, rather than the modelling experiments that followed it, absorbed approximately 90% of the total time invested in this project (Sections 1.3, 3.0), and this work maintains that this allocation of effort was the correct methodological priority for a first Russian-language resource of this kind, on the grounds — argued in Section 1.3 and empirically reinforced by Noticia’s own finding that fine-tuning performance had not saturated even at 750 training examples (Section 2.1.2) — that corpus quality and scale, not model sophistication, constitute the binding constraint on this task.

On modelling (Objectives 6–8). Four general-purpose LLMs were evaluated under a single, unified few-shot protocol applied identically across all four models (Section 4.2), and ClickbaitFighter-7B was adapted to Russian through an explicitly-framed exploratory, resource-constrained fine-tuning experiment (Section 4.3). The resulting evaluation (Chapter 5) found that, among the few-shot models, Qwen-family models outperformed the comparably-sized Llama-3.1-70B-Instruct, echoing a pattern already documented for Spanish in Noticia (Section 5.2.1); and that the exploratory fine-tuning experiment did not, under the automatic metrics available and the severe hardware constraints under which it was conducted, reproduce the fine-tuning advantage Noticia documented for Spanish (Section 5.4) — a result this work attributes principally to the identified, resource-driven artefacts of that specific experiment (Section 5.3.1) rather than to a principled claim about the viability of cross-lingual adaptation in general.

Overall, this work concludes that its central contribution is the corpus itself and the methodology developed to construct, structure, and quality-audit it (Chapters 2–3), while its modelling experiments (Chapters 4–5) are more accurately characterised as a first, resource-constrained empirical probe of that corpus’s usability than as a definitive benchmark of Russian clickbait-resolution capability — a distinction the following section develops into concrete directions for future work.

6.2 Limitations and Future Work

The limitations acknowledged throughout this Memoria are drawn together here, each paired with a concrete, actionable direction for future work.

6.2.1 Corpus scale relative to the NoticIA benchmark

NoticIA’s own analysis of fine-tuning performance as a function of training-set size (Section 2.1.2, Figure 6 in the original paper) found that ROUGE performance had not yet plateaued at 750 training examples, suggesting 700 or more annotated clickbait examples as a practically motivated lower bound for a resolution-training corpus of this kind to approach its performance ceiling. The corpus constructed in this work falls short of that benchmark: 600 annotated clickbait examples in total (Section 3.2.4), reduced to 538 in the class-and-length-filtered `_balanced_ready` variant actually used for model training and evaluation in Chapters 4–5 (Section 3.4.3). This shortfall is a direct, acknowledged consequence of the annotation effort — carried out by a single annotator (Section 3.3.2) within the time budget of a TFM — being the binding constraint on corpus growth, rather than of any limitation in the underlying raw material: as detailed in Section 6.2.3 below, the raw crawled corpus (2,427 articles, Section 3.1.3) contains substantially more material than has so far been annotated, meaning that closing this gap in future work is a matter of continued annotation effort against already-collected data, rather than of further data collection.

6.2.2 A distinguishing strength: balanced detection-oriented design

Where the present corpus falls short of NoticIA on raw clickbait-example count, it offers a complementary strength that NoticIA’s design does not: a **leakage-audited, category-balanced pairing of clickbait and non-clickbait examples** (594 non-clickbait against 600 clickbait rows, Section 3.2.2, Section 3.4.2), collected from an undifferentiated, several-month news stream rather than from a pre-curated clickbait-only source (Section 3.1.1). NoticIA, built specifically and exclusively for the resolution-generation task, offers no non-clickbait counterpart at all, and is therefore not directly usable, without substantial supplementary data collection, for training or evaluating a realistic clickbait *detection* classifier of the kind the balanced corpus variant constructed in this work is explicitly designed to support (Section 3.4.2). This work maintains that this balanced, leakage-audited detection-oriented resource, released alongside the resolution-oriented `_full` corpus, constitutes a distinguishing methodological contribution in its own right, complementary to — rather than merely a smaller replication of — NoticIA’s resolution-only design, and one that future work should build upon by training and formally benchmarking the graded-severity classifier that the taxonomy of Section 2.2 was specifically designed to support, an

experiment outside the scope of the modelling chapters of this TFM (Chapter 4) but directly enabled by the corpus schema constructed in Chapter 3.

6.2.3 Completing annotation of the full raw corpus

The single most concrete, immediately actionable direction for future work identified in this Memoria follows directly from the gap between the raw crawled corpus and the currently annotated subset documented in Chapter 3: of the **2,427** raw articles collected across the four sources (Section 3.1.3), only **1,296** (53.4%) have, to date, been carried through the full manual annotation protocol described in Section 3.3. Completing annotation of the remaining, already-collected but not-yet-annotated articles is a direct extension of exactly the same methodology already validated and applied in this work — requiring no further crawler development (Section 3.1.2), no schema redesign (Section 3.2.4), and no revision of the annotation taxonomy (Section 2.2) — and, based on the class distribution already observed within the currently annotated subset (approximately 46.3% clickbait among annotated rows, Section 3.2.4), extending annotation to the full raw corpus could plausibly yield **in excess of 1,000 annotated clickbait examples paired with a comparably-sized, leakage-audited non-clickbait set** — comfortably exceeding the approximately 700-example benchmark associated with NoticIA’s reported performance ceiling (Section 6.2.1), while simultaneously preserving and substantially strengthening the balanced, detection-oriented design identified as this corpus’s distinguishing strength in Section 6.2.2. To the best of the author’s knowledge, and as argued in Section 1.2, no openly available Russian-language resource of this combined scale, annotation granularity, and balanced detection/ resolution dual-purpose design currently exists; completing this annotation effort is therefore proposed as the highest-priority future extension of this work, directly continuing — rather than departing from — the corpus-first methodological priority argued for throughout this Memoria (Section 1.3, Section 6.1).

Table 6.1: Projected corpus scale under completed annotation

Source	Raw (collected)	Annotated (current)	Remaining (unannotated)
aif.ru	472	472	0
eg.ru	384	383	1
lenta.ru	1,271	369	902
mail.ru	300	72	228
Total	2,427	1,296	1,131

Table 6.1a: Raw, annotated, and remaining article counts per source. aif.ru and eg.ru are effectively fully annotated; the remaining material is concentrated in lenta.ru (902 articles) and mail.ru (228 articles).

Class	Current (annotated)	Projected additional ¹	Projected total
Clickbait	600	+524	1,124
Non-clickbait	696	+607	1,303

Class	Current (annotated)	Projected additional ¹	Projected total
Total	1,296	+1,131	2,427

¹ *Projected by applying the current annotated clickbait rate ($600/1,296 = 46.3\%$) uniformly to the 1,131 remaining raw articles. This is a first-order estimate: the actual clickbait rate among the unannotated material — concentrated in *lenta.ru* and *mail.ru* (Table 6.1a) — may differ from the *aif.ru/eg.ru*-heavy composition of the currently annotated subset.*

Table 6.1b: Projected class-level corpus scale assuming completed annotation of all 2,427 raw articles, comfortably exceeding the ~700-example benchmark associated with Noticia’s reported performance ceiling (Section 6.2.1).

6.2.4 Multi-annotator validation

As discussed at length in Section 3.3.2, the annotation reported in this work was carried out by a single annotator, departing from Noticia’s double-annotation protocol, which allowed its authors to report a Spearman correlation of 0.99 between model rankings computed against two independent annotators’ gold summaries (Section 2.1.2) — a form of validation this work has not been able to replicate. A natural, and comparatively low-cost, extension of the completed-annotation effort proposed in Section 6.2.3 would be to recruit one or more additional annotators — ideally native Russian speakers independently trained on the taxonomy and guidelines developed in Section 2.2 and Section 3.3.1 — to produce a second, independent set of `answer_summary` annotations (and, ideally, independent ten-feature tag assignments) for a representative validation subset of the corpus, enabling both a formal inter-annotator agreement statistic (analogous to Noticia’s Spearman correlation) and a principled empirical check on the taxonomy-driven reliability arguments advanced, but not independently verified, in Section 3.3.2.

6.2.5 Scaling the modelling experiments

Finally, and directly following from the resource-constraint framing of Section 4.1 and Section 4.3.1, the modelling experiments of this work should be understood as a first, deliberately narrow-scoped probe of the constructed corpus rather than an exhaustive evaluation, and future work should extend both the breadth and the rigour of that evaluation as compute and budget allow:

- **A broader model pool.** The four-model few-shot comparison of Section 4.2.1 should be extended to a substantially larger and more diverse set of models — spanning additional families (e.g. DeepSeek, Mistral, further Llama and Qwen variants, and, resources permitting, leading commercial models such as GPT-4-class or Claude-class systems, as Noticia itself included for Spanish, Section 4.1 of the original paper) and a wider range of parameter scales — analogous in breadth to Noticia’s own thirty-plus-model zero-shot sweep (Section 2.1.2).
- **Multiple-sampling evaluation.** As acknowledged in Section 4.1, the single-pass, near-deterministic evaluation protocol adopted in this work should be replaced, where budget allows, with a multiple-sampling protocol — generating several stochastic

completions per test instance at a non-trivial decoding temperature and averaging the resulting metrics — to obtain a statistically more robust estimate of each model’s expected performance, rather than a single-draw snapshot.

- **A properly-resourced fine-tuning replication.** Directly following from the diagnostic discussion of Section 5.3.1 and Section 5.4, the cross-lingual fine-tuning experiment of Section 4.3 should be repeated under a substantially larger compute budget — ideally approaching the original ClickbaitFighter training configuration (full-parameter fine-tuning, 8,192-token sequence length, multi-GPU training, Section 2.1.3) — both to test whether the fine-tuning advantage documented for Spanish can be reproduced for Russian once the resource-driven confounds identified in this work (truncated sequence length, 4-bit quantisation, residual output-length inflation) are removed, and to train and evaluate all three ClickbaitFighter scales (2B, 7B, and 10B, Section 2.1.3) rather than the single 7B variant evaluated here, so as to reproduce, for Russian, Noticia’s own finding that even the smallest fine-tuned task-specific model outperformed every zero-shot general-purpose model evaluated (Section 2.1.3).
- **Additional prompting methodologies.** Beyond the single-demonstration few-shot protocol evaluated in Chapter 4, future work should explore multi-shot prompting (conditional on context-length and cost budgets permitting, Section 2.3.3), chain-of-thought prompting, and — directly enabled by the balanced, detection-oriented corpus design discussed in Section 6.2.2 — a joint detection-and-resolution prompting protocol, in which a model is asked to first classify a headline as clickbait or not before attempting resolution, more closely mirroring the two-stage judgement process followed by the human annotator in this work (Section 3.3.1).

Taken together, these five directions define a natural, incremental research programme that builds directly on the corpus and methodology developed in this TFM, rather than requiring a departure from it: each extends a specific, explicitly identified limitation of the present work into a concrete next step, consistent with this Memoria’s central argument that the durable contribution of this project is the corpus itself, and that its value will compound as future work — by this author or by others, given the corpus’s explicitly interoperable design (Section 3.2.4) — continues to build upon it.

The corpus, annotation guidelines, and full experimental code are publicly available at: <https://github.com/Andromeda-git/TFM-project>

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